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FACULTY OF ENGINEERING
COMPUTER ENGINEERING DEPARTMENT

Project Report

CENG 408

Innovative System Design and Development II

**VR Anatomy - VR Based Educational Interactive Human
Anatomy Training Platform**

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Abstract

VR Anatomy is a virtual reality-based educational application designed to support introductory human anatomy learning through interactive 3D visualization, guided exploration, AI-supported explanations, and quiz-based assessment. The system focuses on selected topics from the musculoskeletal and circulatory systems, allowing students to examine anatomical structures in an immersive VR environment.

The application includes four main modules: AI Guided Learning, Free Explore, AI Chat, and Quiz. These modules support structured learning, independent model inspection, question-answer interaction, and knowledge evaluation. The system was developed in Unity, while AI features were supported by a Python-based backend using Retrieval-Augmented Generation, speech-to-text, and text-to-speech technologies.

A preliminary evaluation was conducted with high school students under the neutral interaction condition. The results showed an improvement between pre-test and post-test scores, suggesting that VR Anatomy may support short-term understanding of anatomy topics. Overall, the project demonstrates how VR and AI can be combined to create a more interactive, visual, and supportive anatomy learning experience.

1. Introduction

Human anatomy is a fundamental subject in health education because students need to understand not only the names of anatomical structures, but also their locations, shapes, and relationships with surrounding parts. However, traditional learning materials such as textbooks, diagrams, and classroom presentations may not always be enough to help students develop a clear three-dimensional understanding. This can make anatomy difficult to learn, especially for students who are encountering these concepts for the first time.

Virtual Reality provides an effective way to support anatomy education by allowing students to examine anatomical structures in an immersive and interactive environment. Instead of only viewing two-dimensional images, students can observe 3D models from different angles, interact with selected structures, and explore the body in a more concrete way. This can improve spatial understanding and make the learning process more engaging.

VR Anatomy was developed as a graduation project to support anatomy learning through VR-based visualization and AI-assisted interaction. The current version focuses on selected topics from the musculoskeletal and circulatory systems. These systems were chosen because structures such as bones, muscles, the heart, and vessels benefit strongly from three-dimensional representation.

The application includes four main modules. AI Guided Learning provides structured explanations with avatar support. Free Explore allows students to inspect anatomical models independently. AI Chat enables students to ask anatomy-related questions through text or voice. Quiz allows students to evaluate their understanding after the learning process.

Technically, the project combines a Unity-based VR application with a Python-based AI backend. The backend uses Retrieval-Augmented Generation to provide answers based on prepared anatomy content and supports speech-to-text and text-to-speech features. The project also includes

an experimental design comparing a neutral interaction condition with a planned personalized interaction condition.

This report presents the project work plan, system architecture, implementation details, testing process, evaluation results, and future work of the VR Anatomy system.

2. Project Work Plan

	16.02.2026	27.03.2026	10.04.2026	17.04.2026	24.04.2026	04.05.2026	13.05.2026	15.05.2026	22.05.2026
System Architecture & Design Decisions									
Implementation Details									
Project Poster									
Testing & Validation									
Midterm Demo / GitHub First Release									
Conclusion & Future Work									
Project Development									
Student Testing & Experimental Evaluation									

Figure 1. Project Work Plan for “VR Anatomy - VR Based Educational Interactive Human Anatomy Training Platform”

3. System Architecture & Design Decisions

3.1. Overview

3.1.1. Purpose

The purpose of this report is to document the system architecture and the key design decisions of VR Anatomy. It explains how the Unity VR client, the packaged learning content, and the AI Tutor backend work together. It also states the reasons behind major choices such as scene organization, content storage format, and the separation between learning modes.

3.1.2. System Summary

VR Anatomy is a single user VR application designed for classroom guided use with one headset. The student is the primary user who interacts inside VR. The instructor is a secondary user who guides verbally and does not interact with the application directly.

Core features focus on module based anatomy exploration and a quiz mode that can be started from the main menu.

Main user capabilities are listed below.

- Select a module from the VR main menu

- Explore a 3D anatomy model in VR
- Select an anatomical structure and see it highlighted
- Perform basic interactions such as grab, rotate, and release
- View labels and short info cards for selected structures
- Encounter checkpoint nodes during the “Learn Mode” where short assessment questions are presented
- Answer mini quiz questions at these checkpoints to reinforce understanding of the currently explored anatomical content
- Receive immediate feedback at checkpoint interactions to support active learning
- Start Quiz Mode from the main menu independent from modules
- Start Review Mode where only selected model assets are shown and info cards are not shown
- The AI Tutor provides a dedicated chat and voice interface that is opened from the main menu via the “AI Tutor” button.
- The student can ask questions either by typing or by using push-to-talk (speech-to-text).
- The backend returns a short text answer grounded in the project’s learning content using a Retrieval-Augmented Generation (RAG) approach.
- The answer can optionally be read aloud via a speaker button (text-to-speech).
- In this version, the AI Tutor is not structure-aware and does not automatically use the currently selected anatomical structure as context.
- If the AI service is unavailable, the AI Tutor feature is disabled while core VR exploration and quiz features continue.

3.1.3. Architecture Goals

The architecture was designed to support an educational VR experience that is stable in classroom use and easy to expand with new modules and content. The system includes an AI Tutor feature that may become unavailable due to network or service issues. For this reason, the architecture separates the VR learning functions from the AI Tutor integration.

The main goals are listed below.

- Keep interactive exploration, Quiz Mode, and Review Mode clearly separated so that each mode remains simple and consistent
- Keep core VR functions usable when the AI Tutor service is unavailable, such as module selection, 3D exploration, structure highlighting, and basic interactions
- Allow content updates by changing JSON files and Unity model assets without changing the application logic
- Keep runtime behavior predictable for a single headset and a single student session
- Make the flow of information clear between user interface, content loading, interaction handling, and AI Tutor requests to simplify testing and troubleshooting
- Reduce reliance on online services so that non AI functions can still run in offline classroom conditions

3.2. High Level Architecture

This section describes the main building blocks of VR Anatomy and how data moves between them during key user actions. The system is centered on a Unity-based VR client that loads local content for learning and communicates with a cloud-based AI backend powered by Gemini for the AI Tutor feature when available.

3.2.1. Architecture Diagram and Explanation

The diagram below shows the high-level components and their connections. The VR client runs on the headset, loads JSON-based educational content and 3D assets locally, and communicates with the AI backend only for AI Tutor chat and voice requests.



Figure 2. High-level system architecture of the VR Anatomy.

The architecture follows a hybrid approach with a clear separation between local learning functionalities and cloud-based AI services.

The VR learning features are fully supported by local resources, allowing the application to operate without network connectivity. This ensures uninterrupted access to core functionalities such as module exploration, checkpoint interactions, and quiz modes.

The AI Tutor feature, on the other hand, relies on an external backend integrated with the Gemini API. User queries are sent to the backend, where a Retrieval-Augmented Generation (RAG) pipeline processes the request and returns a context-aware response. If the AI service becomes unavailable due to network issues or API limitations, the AI Tutor feature is temporarily disabled, while all core VR functionalities continue to operate normally without interruption.

3.2.2. Main Components

Unity VR Client:

The VR client is the primary runtime environment. It renders the 3D anatomy scenes, handles XR input (controllers/hands), and displays all in-app UI. The client provides the main user modes: AI Guided Learning, Free Review, Quiz, Settings/About, and a dedicated AI Tutor screen. The VR client is designed to function offline for core exploration and quiz features.

Main Menu and Global Screens:

The main menu is the entry point for navigation across the application. From this screen, the student can select modules for learning/review, start the quiz mode, open settings and the about page, or access the AI Tutor. The AI Tutor is launched from the main menu via the “AI Tutor” button.

Scene and Mode Management:

A scene/mode management layer is responsible for launching the appropriate Unity scene and configuring the active mode. When the student starts a module, the system loads the required 3D assets and prepares the UI for the selected mode (learning vs. review). This separation prevents quiz and AI features from coupling tightly with the main exploration flow.

Interaction and Selection Handling:

The interaction subsystem manages core VR interactions such as grabbing, rotating, releasing, and selecting anatomical structures. It also triggers visual feedback (e.g., highlight) when a structure is selected. The selection output is used by the UI to show labels and short info cards, and by content lookup logic to fetch the correct text entry.

Local Content Layer (Assets + JSON):

The application ships with a local content package that includes 3D model assets and JSON files. These JSON files contain structure labels/descriptions and the quiz question pool. The exported JSON is the runtime source of truth for learning text and quizzes, enabling offline use and keeping content changes independent from Unity scripts.

Content Loader:

A content loader reads the packaged JSON data and serves the appropriate label and short description for a selected structure. The loader uses stable identifiers/tags (e.g., `module_tag` for scope and `structure_id` for individual objects) to perform a simple lookup and return the correct UI text.

Quiz Manager:

The quiz manager loads questions from the local JSON pool, displays them in VR, processes student answers, and shows immediate feedback and results. Quiz mode is intentionally independent from module exploration, allowing it to be launched directly from the main menu without requiring a module to be active.

AI Tutor Client:

The AI Tutor client is a dedicated UI screen that allows the student to ask questions via typing or push-to-talk (speech-to-text). It sends the question to the backend and displays a short text answer. Optionally, the student can play the answer aloud using text-to-speech. In the current scope, the AI Tutor is not structure-aware and does not automatically incorporate the currently selected structure as context.

Gemini-Based AI Backend (RAG Service):

The backend receives AI Tutor questions and generates grounded responses using a Retrieval-Augmented Generation (RAG) approach integrated with the Gemini API.

When a user submits a query, the system first retrieves relevant content snippets from the knowledge store. These retrieved documents are then provided as contextual input to the Gemini model, which generates a concise and context-aware response aligned with the project's educational material.

This design ensures that responses are not purely generative, but are grounded in verified learning content, reducing hallucination and increasing reliability.

If the backend or Gemini service becomes unavailable, the AI Tutor feature is disabled, while core VR exploration, checkpoint interactions, and quiz features remain fully functional.

Retrieval Index / Knowledge Store:

The retrieval layer stores structured learning content (such as curated notes, anatomical descriptions, and instructional material) in a format optimized for semantic search.

This layer is responsible for identifying and returning the most relevant text chunks based on the user's query. These retrieved segments are then passed to the RAG pipeline as contextual input for the Gemini model.

By separating retrieval and generation, the system ensures that answers remain consistent with the predefined educational content while still benefiting from the flexibility of a large language model.

3.2.3. Key Data Flows

This subsection summarizes the main runtime flows of VR Anatomy. Each flow is explained from the initial user action to the visible outcome inside the headset. For each flow, an activity diagram is provided to show the control steps and the decision points.

3.2.3.1. Module Launch Flow

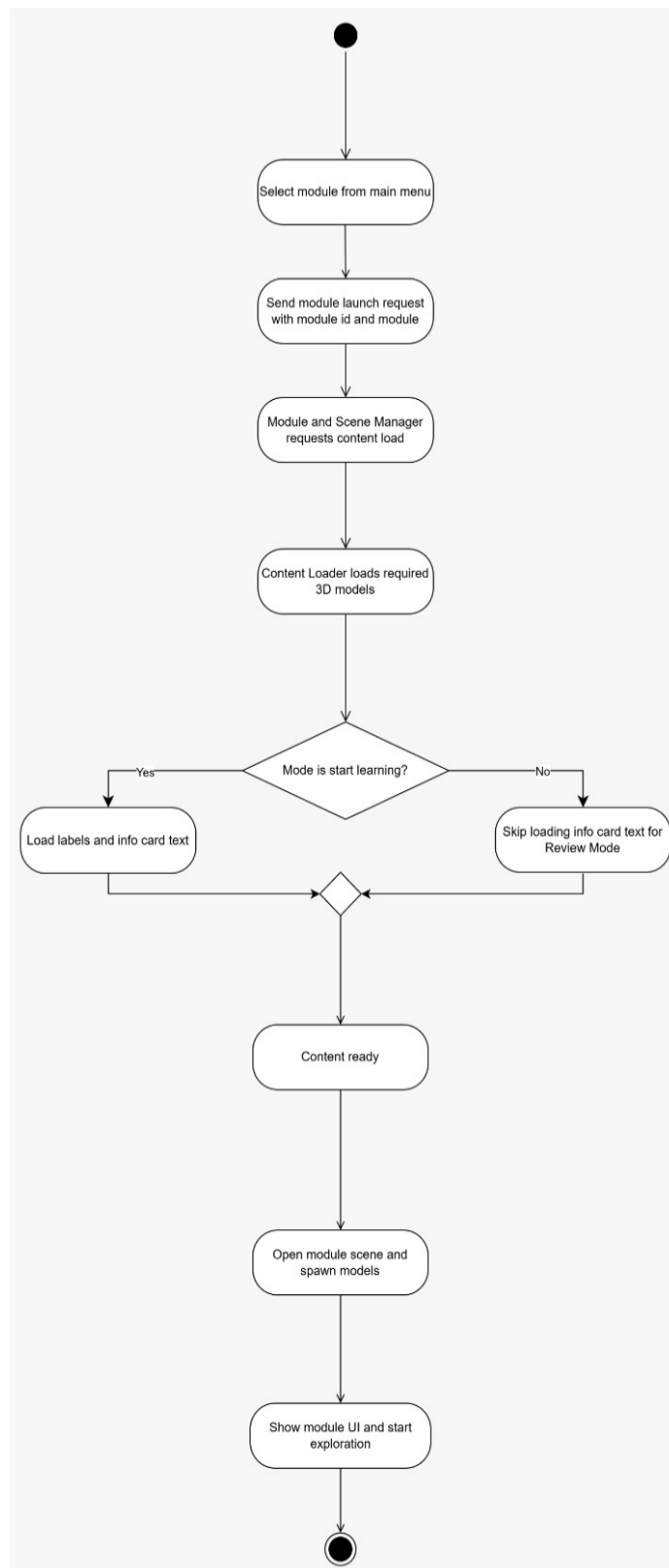


Figure 3. Activity diagram of the “Module Launch Flow”.

3.2.3.2. Structure Selection and Info Card Flow



Figure 4. Activity diagram of the “Structure Selection and Info Card Flow”.

3.2.3.3. Quiz Mode Flow

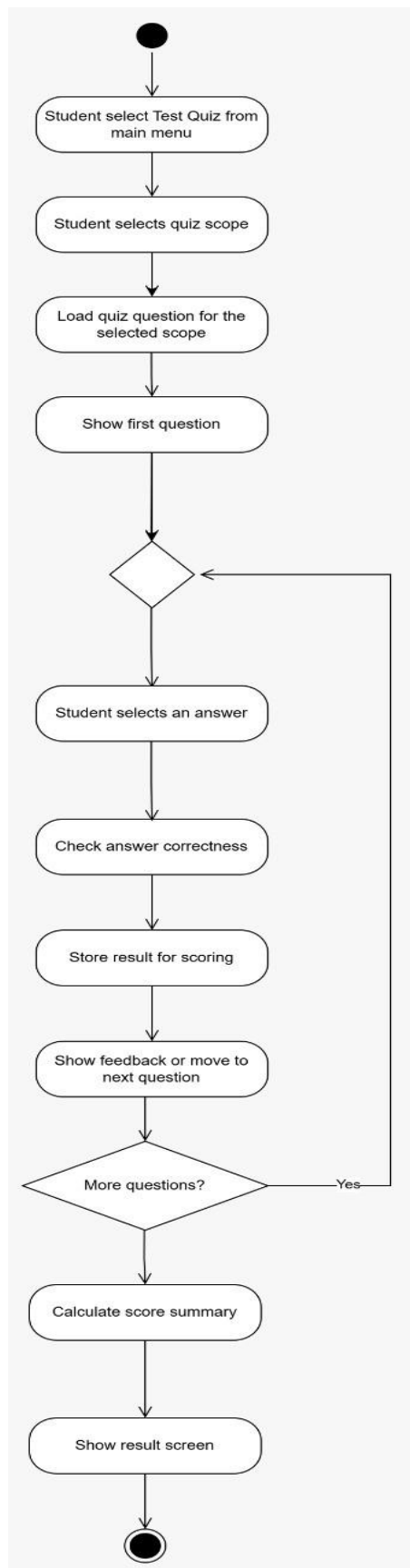


Figure 5. Activity diagram of the “Quiz Mode Flow”.

3.2.3.4. Review Mode Flow

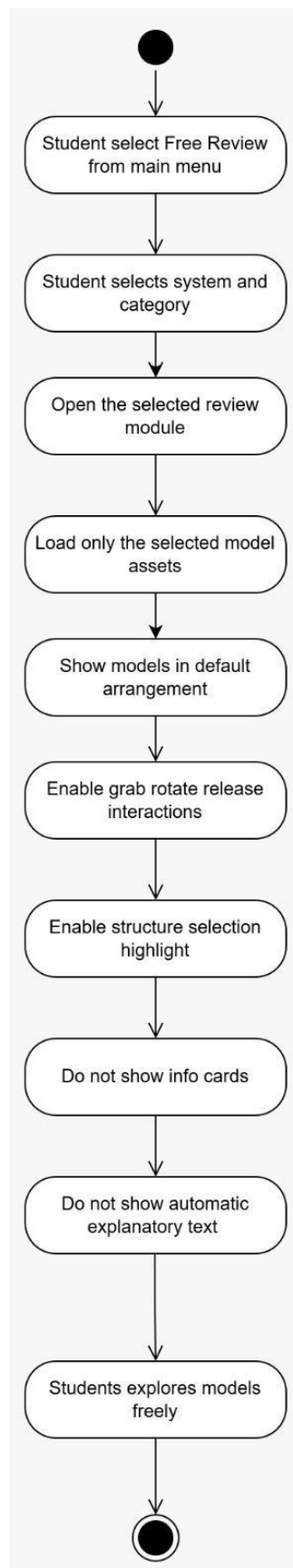


Figure 6. Activity diagram of the “Review Mode Flow”.

3.2.3.5. AI Tutor Chat Flow

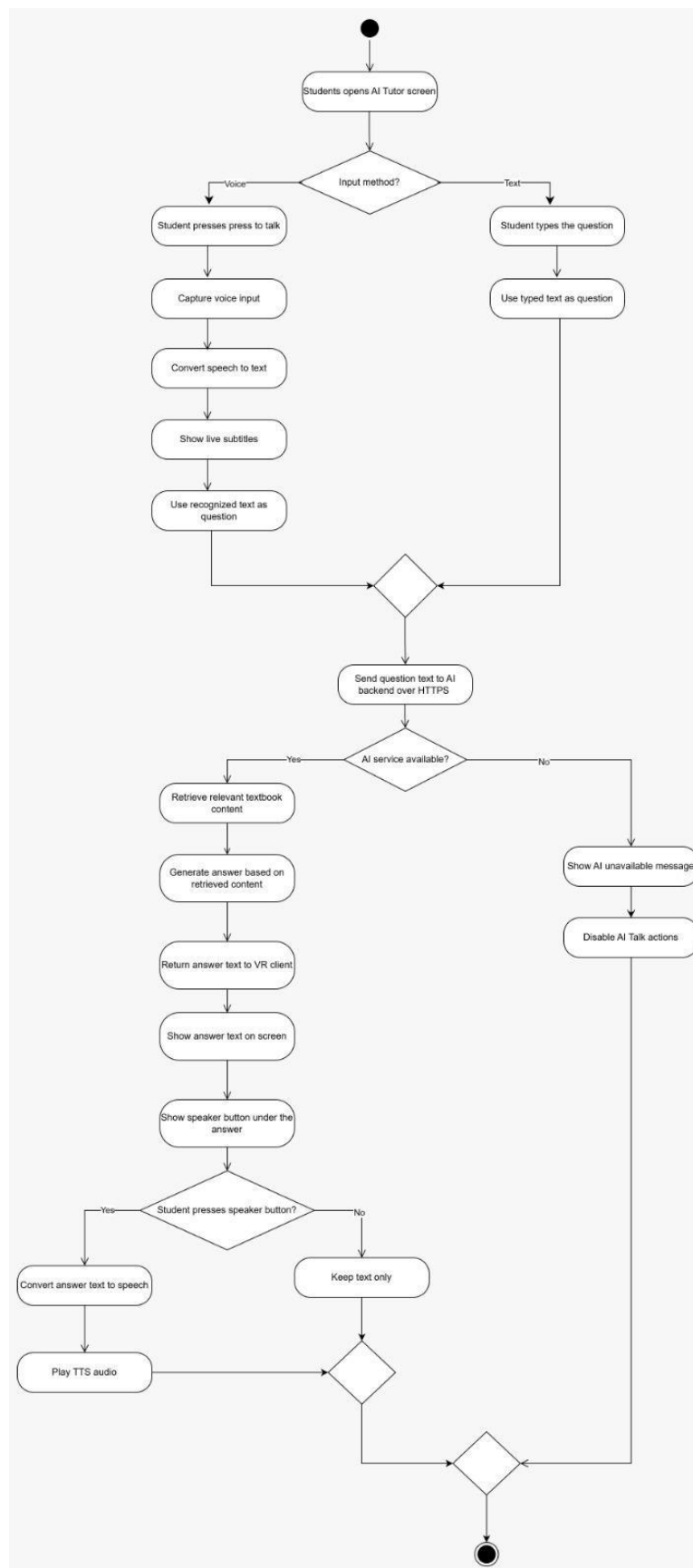


Figure 7. Activity diagram of the "AI Tutor Chat Flow".

3.3. Module Design in Unity.

This section explains how VR Anatomy is organized inside Unity with the current main menu structure. It describes scene and mode organization, interaction handling, user interface design, content mapping, and fallback behaviors that keep the experience usable

3.3.1. Scene and Module Structure

The Unity project is organized around a main menu entry scene and a small set of mode scenes or mode states. The main menu provides access to learning, free review, quiz, settings, AI tutor, and about screens.

A practical scene and mode structure is listed below:

- Main Menu scene as the entry point
- Learning mode scenes for module based exploration
- Free Review mode scenes that load selected model assets without info cards
- Quiz mode scene or UI flow that is independent from learning and free review modes
- AI Tutor scene that opens as a full screen chat and voice interface
- Settings scene or panel that is reachable from the main menu
- About scene that shows project summary and credits

The main menu includes a hierarchical navigation structure for learning and free review. This navigation determines which content and model assets are loaded.

AI Guided Learning supports guided exploration using hierarchical category selection. The student first selects an anatomy system and then selects a category within that system. In the current scope, both the Musculoskeletal System and the Circulatory System follow the same menu structure.

In Free Review mode, only the selected model assets are loaded, and explanatory elements such as info cards and automatic descriptive texts are disabled to provide a distraction-free exploration experience. However, the system offers an optional visual aid that allows the user to display the name of the currently held anatomical structure. This feature can be toggled on or off by the user.

Test Quiz is accessed from the main menu and is independent from the category navigation used in learning and review. It provides four entry options for question pools. Musculoskeletal System, Circulatory System, Basic Concepts and All Systems.

Settings, AI Tutor, and About are standalone screens that are reached directly from the main menu. They do not use the anatomy category hierarchy.

3.3.2. Interaction Layer

Key responsibilities are listed below:

- Detect structure selection using ray targeting or direct hand targeting
- Apply visual highlight to the selected structure
- Trigger label display and info card display in learning mode
- Support grab, rotate, and release interactions
- Provide a reset action on the controller that clears the current selection and restores the module to its initial state

The reset action supports classroom use. After the student moves or inspects parts, a single button press returns the model arrangement to the default state seen when the module first opened. This reduces confusion and helps the instructor quickly continue the lesson.

3.3.3. UI Layer

The UI layer supports navigation and learning feedback inside VR. It includes the main menu, mode entry screens, and in scene feedback such as labels and short info cards. The UI is designed to remain readable and to avoid covering the 3D model during exploration.

The UI is organized into four parts. These are:

- Main menu and global screens
- In module UI for learning and free review
- Quiz UI
- AI Tutor UI

3.3.3.1. Main Menu and Global Screens

The main menu is the entry point of the application. It provides access to AI Guided Learning, Free Review, Test Quiz, Settings, AI Tutor, and About. AI Guided Learning and Free Review use hierarchical category selection to determine which module content and model assets are loaded.

Settings is a global screen intended for configuration such as sound related options. The exact set of settings may be expanded later. About is a global screen that presents project summary, team information, contributions and versions.

3.3.3.2. In-Module UI for Learning and Free Review

The in-module UI supports exploration while the student interacts with 3D models.

In Learning Mode, the system provides guided interaction by displaying labels and short informational cards for the selected anatomical structure. These elements help the student understand the content during exploration.

Additionally, Learning Mode includes interactive checkpoint elements placed at specific points within the module. At these checkpoints, the system presents short assessment questions related to the currently explored anatomical structures. The student can respond to these mini questions and receive immediate feedback, reinforcing learning during the exploration process.

Free Review follows the same exploration flow but removes instructional elements such as info cards and automatic explanatory text to create a simplified and distraction-free environment. However, an optional label can be enabled to display the name of the currently held structure, without additional descriptive content.

The in-module UI elements are listed below:

- A label that shows the selected structure name
- An optional label in Free Review mode that displays the name of the currently held structure
- A short info card in Learning Mode that shows a brief description
- Interactive checkpoint prompts in Learning Mode that present mini questions
- Immediate feedback indicators for checkpoint responses
- A minimal navigation element to return to the main menu
- A reset control that restores the module to its initial state

3.3.3.3. Quiz UI

Quiz UI is separate from modules and is accessed from the main menu. It is designed for clear question presentation, quick answering, and continuous learning support.

Quiz UI elements are listed below:

- Question text area
- Answer options with selection feedback
- Result feedback such as correct or incorrect
- Controls for next question and exit to the main menu
- A contextual hint button that appears if the student does not answer within a certain time

Additionally, the system provides a post-quiz analysis screen after all questions are completed. This screen evaluates the student's performance and identifies topics with a high number of incorrect answers.

Based on this analysis, the system recommends that the student revisit specific topics

by redirecting them to the corresponding sections in the “AI Guided Learning” mode. This creates a feedback loop between assessment and learning, supporting a more adaptive and guided educational experience.

3.3.3.4. AI Tutor UI

AI Tutor opens as a dedicated screen from the main menu. It supports voice-based interaction while showing text to keep answers clear in a classroom environment.

AI Tutor UI elements are listed below:

- Press-to-talk input control
- Live subtitles that show speech recognition output
- A control to play the answer using text-to-speech when available
- A contextual unavailable state message shown only when the AI Tutor feature is actively used and the service cannot be reached

The system follows a context-aware feedback approach for service availability. If the internet connection is lost while the user is inside the AI Tutor screen, a short message is displayed and input controls are disabled.

However, if the application is in other parts of the system (such as the main menu or learning modules), no warning message is shown even if there is no internet connection, since core VR functionalities do not depend on the AI service. Only the AI Tutor button is deactivated in the background.

This prevents unnecessary interruptions and ensures a clean user experience. Core VR features remain fully usable regardless of AI service availability.

3.3.4. Content Mapping

At runtime, the mapping follows a clear sequence.

- The student selects a category from the main menu
- The VR client loads the corresponding module content and model assets
- The content loader reads the JSON entries for that `module_tag`
- When the student selects a structure, the VR client uses `structure_id` to request label and description data
- The UI layer shows the label and short info card in Learning Mode

Free Review uses the same module and structure identifiers but does not display full info cards. The mapping still supports selection highlighting and label display.

In this mode, the system can optionally display the name of the selected or currently held structure. This name is retrieved from the same JSON entries used for info cards; however, only the label field is used, and the descriptive content is intentionally omitted.

Quiz mode uses a different mapping entry point. The student chooses a question pool option in the quiz screen. The VR client then loads the related pool from JSON. The All Questions option merges or selects a combined pool defined in the content package.

Module and structure identifiers are assigned and maintained during content preparation. The same identifiers are stored in JSON and in Unity objects, which keeps content mapping consistent for the current project scope.

3.3.5. Error Handling and Fallback Behaviors

The Unity client should remain usable when some content is missing or when external services are unavailable. Fallback behaviors are designed to keep the student in a valid learning state and to avoid blocking core exploration.

Error handling follows two principles:

- Fail safely by disabling only the affected feature
- Communicate clearly using short, non-technical messages in VR

Typical cases and expected behaviors are listed below:

- If a structure has no JSON description, the client still allows selection and highlighting, and shows only the label
- If a module content package is incomplete, the client blocks entry to that category and returns to the main menu with a short message
- If a quiz pool is missing for a selected option, the client disables that option or shows a message and returns to the quiz selection screen
- If the AI service is unreachable, the client shows an unavailable state and disables AI input, while learning, free review, and quiz remain usable
- If the AI response is slow, the client shows a loading state and applies a timeout, then shows a short message
- If speech recognition or text to speech is unavailable, the client keeps the text based interaction when possible and shows that voice features are unavailable

During development, errors should be logged consistently so the team can reproduce issues. During classroom use, messages should remain short and focused on the next action the student can take.

3.4. API Design

This section describes the minimal API between the Unity VR client and the AI backend powered by Gemini. The API is used only for the AI Tutor feature. Core VR features such as module exploration, selection highlighting, info cards, and quiz use local content and do not

require the API.

The API is designed to be simple, stable, and easy to test. It uses HTTP over TLS and JSON messages. The client sends a question with module context, and the backend returns a grounded answer based on project provided learning content.

3.4.1. API Overview

The AI Tutor feature uses one endpoint. This endpoint accepts a question and returns a grounded answer. The backend uses a retrieval step over project provided learning content and then produces a response based on the retrieved text. This is the RAG approach in this project. It helps the system answer using the provided material instead of guessing from general knowledge. The AI Tutor is not structure aware. The request does not include selected structure fields.

3.4.2. AI Tutor Endpoints

The AI Tutor feature uses a minimal REST endpoint.

The Unity client calls this endpoint directly once the integration is complete. The Gemini-based RAG system operates by using the project's learning content as its primary source.

The purpose of the endpoint is to accept a student's question and return a response based on the project's learning content. The generated responses are sent directly to the Unity client.

AI Tutor does not have structural awareness. The request does not include selected structural information.

3.4.2.1. Request Schema

The request body is JSON. The schema is described as a planned contract for the Unity client.

The only strictly required input for answering is the student question. Additional fields may be added to support integration, debugging, and content scoping.

Recommended minimal field is listed below:

- question as a string and required to generate an answer

Recommended fields that may be used during Unity integration are listed below:

- module_tag as a string to limit retrieval to a selected scope if the content is later split by systems or categories

Design notes are listed below:

- If one textbook is used as a single content source, module_tag may be omitted
- If multiple content scopes are introduced later, module_tag helps keep answers

relevant

3.4.2.2. Response Schema

The response body is JSON. The schema is described as a planned contract for the Unity client because Unity integration is a later step.

The main goal of the response is to return a short answer that can be displayed comfortably in VR. The response may also include supporting references when the backend can provide them.

Recommended minimal fields are listed below:

- answer as a string

Optional design notes are listed below:

- citations are useful to increase trust because they show a brief supporting excerpt from the learning content
- if citations are not available in the current backend setup, the response may contain only answer

3.4.2.3. Error Responses and Timeouts

The AI Tutor endpoint must fail safely because the core VR learning experience should remain usable even when the online service is unavailable. Error handling is designed to be predictable for the client and understandable for the user.

General principles:

- The VR client treats the AI Tutor as an optional feature.
- When an AI request fails, the client keeps the current module state unchanged.
- Errors are logged for debugging, but logs must avoid personal data.

Planned error response shape:

- error_code as a short stable identifier
- message as a user safe explanation

Expected error categories:

- Invalid request: Returned when required fields are missing or values are not valid.
- Authentication or authorization failure: Returned if the endpoint later requires keys or tokens.
- Upstream AI failure: Returned when the hosted AI pipeline cannot produce an answer due to temporary issues.
- Connection timeout is short to detect offline states quickly.
- Total request timeout is moderate to allow retrieval and generation, but still protects interactivity.
- If a timeout happens, the client may allow a manual retry rather than automatic repeated retries.

Client behavior on failure:

- Disable or gray out AI Tutor entry points when the service is unreachable.
- Keep main menu, module browsing, selection, labels, and quiz features available.
- Avoid repeated popups. Prefer a single banner style message and an explicit retry action.

3.4.3. Security and Privacy Considerations

VR Anatomy is a single user VR application with an optional online AI Tutor. Security and privacy work focuses on reducing risk while keeping the system simple for an academic project.

Data minimization:

- The AI request should include only what is needed to answer the question.
- The system should not send selected structure identifiers because the AI Tutor is not structure aware.
- The system should avoid sending user identifiers.

Personal data handling:

- The VR application does not need personal profiles or accounts for the current scope.
- If speech features are enabled in AI Tutor, audio should be treated as transient input and not stored by the VR client.

Transport and endpoint protection:

- If an API key or token is required in the future, it should not be hard coded in the Unity client.

Logging and diagnostics:

- Logs should help debugging without exposing user content.
- By default, logs should not store full questions or audio transcripts.
- If limited logging is needed during testing, it should be time bounded and access controlled.

Abuse and reliability:

- The client should implement safe retries with backoff and stop after a small number of attempts.
- When AI is unavailable, the client should disable AI Tutor while keeping core learning and quiz features available.

Content safety boundaries:

- The AI Tutor is intended for anatomy learning questions that relate to the provided textbook.
- The UI should set expectations that answers may be incomplete or wrong.
- The system should encourage students to verify critical information with course material or an instructor.

3.5. Deployment and Runtime Environment

This section describes how VR Anatomy runs during a typical session and how the optional AI Tutor fits into the runtime. The goal is a stable classroom style setup with a single headset and a single student at a time.

3.5.1. Runtime Setup and Execution Context

VR Anatomy runs as a Unity-based VR application on a single headset. One student uses the headset at a time. The instructor is a secondary user who guides verbally and does not interact with the application directly.

Runtime characteristics:

- The core learning experience runs locally on the headset.
- Learning modules load 3D models from Unity assets.
- Labels and short info cards use local text content.
- Review Mode uses the same model assets but does not show info cards or automatic explanatory text; however, an optional label can be enabled to display the name of the currently held structure.
- Quiz Mode is started from the main menu and runs independently from learning modules.

The AI Tutor is an optional feature accessed through the AI Tutor screen. When available, it connects to a Gemini-powered backend over the network. If the AI service cannot be reached, AI Tutor is disabled while the rest of the VR application remains usable.

3.5.2. Offline and Online Operation

VR Anatomy is designed so that learning modules work without online services. Online access only improves the experience through AI Tutor and enhanced interactive features.

Offline operation:

- AI Guided Learning modules are available with 3D exploration and basic interactions.
- Selecting a structure still highlights it.
- Labels and short info cards still appear for selected structures.

Review Mode works and keeps its rule set with no info cards or automatic text; an optional label can still display the structure name.

- Quiz Mode works as long as quiz data is available locally for the build used in testing.
- Checkpoint interactions remain available; however, only predefined multiple-choice questions are presented when there is no internet connection.

Online operation:

- All offline features remain available.
- AI Tutor becomes available when the AI backend can be reached.
- Checkpoint interactions are enhanced with AI-supported question types, such as dynamic or voice-based questions. These questions may utilize a RAG-based pipeline in the backend to generate or evaluate responses.
- If the AI service fails during a session, AI Tutor and AI-supported checkpoint questions are disabled, while the rest of the application continues to function normally.

3.5.3. Dependencies and Configuration

The project uses a small set of dependencies to keep the build stable and easier to test.

VR client dependencies:

- Unity project with VR interaction support
- XR plugins and device runtime required by the selected headset
- 3D model assets included as Unity assets
- UI assets for menus, labels, info cards, and quiz screens

AI Tutor dependencies:

- A Gemini-based backend that exposes the AI answer endpoint
- A retrieval setup that uses the indexed textbook content
- A language model service used by the backend

Configuration approach:

- VR client settings are stored in Unity project configuration and prefabs.
- Secrets such as keys should not be stored in the VR client. They should be handled on the server side.
- Quiz and structure descriptions are exported to JSON.

3.6. Design Decisions

This section explains the main design choices behind VR Anatomy. Decisions were made to support a stable VR learning experience, limit development risk, and keep the system testable within a graduation project scope.

The project prioritizes:

- Reliable core learning features that work without online services
- Simple interactions that students can learn quickly
- A modular structure that supports iterative content growth
- Clear separation between VR learning and optional AI support

3.6.1. Why Module-Based Scenes

The application uses module based scenes to keep content isolated and easier to manage.

Key reasons:

- Each module can be loaded and tested independently.
- Large 3D assets stay inside the relevant module and do not affect other modules.
- Scene separation supports parallel work. One team can work on UI while another team improves a specific module.
- Memory usage is easier to control because only the active module needs to be loaded.
- The reset button behavior is easier to implement. Reset returns the module to the initial state by clearing selection and restoring default arrangement.

3.6.2. Why JSON for Content

The project stores learning text and quiz content in JSON files to separate content from Unity code. These JSON files are bundled with the VR application and serve as the runtime source of truth for labels, short descriptions, and the quiz pool.

Expected benefits:

- Content can be updated without changing Unity scripts.
- Structure descriptions and quiz questions follow a consistent format.
- Core learning and quiz features can work offline using the packaged JSON content.

Status and limits:

- During development, quiz questions may be authored in an Excel file for convenience, but the VR client uses exported JSON at runtime.
- JSON files are shipped with the app as part of the local content package.
- 3D models remain Unity assets and are not part of the JSON content.

3.6.3. Why Review Mode Without Info Cards

Review Mode removes info cards and automatic explanatory text to support focused visual learning.

Rationale:

- It encourages students to rely on recognition and exploration rather than reading.
- It supports instructor guided sessions where explanations are given verbally.
- It reduces visual clutter and keeps the scene clean for examination.
- It separates learning from assessment. The same model can be reviewed without hints.

Resulting behavior:

- Only the selected model assets are shown.
- Selection highlighting and basic grab and rotate interactions remain available.
- No info cards appear after selection.
- An optional label can be enabled to display the name of the currently held structure without additional descriptive content.

3.6.4. Why AI Tutor Is Optional and Not Structure-Aware

The AI Tutor is optional because the core VR experience must remain usable offline and without external dependencies. The AI pipeline currently works directly in Unity.

Reasons for optional integration:

- Classroom use must not fail when the network is unstable.
- Testing and demos remain possible without relying on online services.
- The system scope stays realistic for a graduation project.

Reasons for not being structure aware:

- The current AI setup is based on the textbook ingestion and retrieval flow.
- Passing selected structure identifiers would suggest precision that the system does not reliably support.
- Keeping requests minimal reduces coupling between VR interaction state and the AI backend.

User experience rule:

- If the AI service is not available, AI Tutor is disabled while modules and quiz remain available.

3.6.5. Performance and VR Comfort Decisions

VR comfort and performance are treated as core quality goals because the application runs on a headset and uses real time interaction.

Performance decisions:

- Keep scenes simple and avoid loading unnecessary assets outside the active module.
- Prefer optimized 3D models suitable for real time VR.
- Use stable lighting choices where possible and avoid expensive effects during interaction.
- Limit the number of on screen UI elements during exploration.

Comfort decisions:

- Avoid forced camera movement and sudden viewpoint changes.
- Prefer direct object interaction such as grab, rotate, release.
- Provide the reset button to quickly return to a known safe state.
- Keep UI readable and consistent in size and placement.

3.7. Limitations and Risks

This section lists the main limitations and risks of VR Anatomy within the current project scope. The goal is to make constraints explicit and define what may affect stability, learning and quality.

3.7.1. Technical Limitations

The project targets a single headset and a single user at a time. This keeps the system simple but limits usage scenarios.

Main limitations:

- No multiuser support and no shared session features.
- No instructor control panel. The instructor guides verbally outside the system.
- Limited hardware coverage. The project is tested on the chosen headset and configuration.
- Module content depends on Unity assets. Large assets may increase load times and memory use.

Practical impact:

- The application is suitable for guided demos and small classroom sessions.
- Porting to other devices may require extra XR configuration and performance tuning.

3.7.2. AI-Related Risks

The AI Tutor improves access to explanations, but it introduces risks because it depends on online services and generated text.

Main risks:

- Availability risk. Network or backend issues can disable AI Tutor.
- Answer quality risk. Generated answers may be incorrect, incomplete, or misleading.
- Context limitation. The AI Tutor is not structure aware, so it may not understand what the student is looking at in VR.
- Safety and privacy risk. Voice input may include personal data if users speak freely.

Mitigation approach:

- Keep AI optional and ensure VR modules and quiz work without it.
- Keep requests minimal and avoid sending structure identifiers.
- Avoid storing voice or question content in logs by default.

3.8. Conclusion

VR Anatomy is a Unity based VR learning application that supports interactive anatomy exploration for a single student using a single headset. The system provides module selection from a main menu, 3D exploration, structure highlighting, basic grab and rotate interactions, labels, and short info cards. Quiz Mode is available independently from modules, while Review Mode offers a clean study view without info cards or automatic explanatory text. A reset action restores each module to its initial state to support repeatable practice.

The architecture cleanly separates core VR learning features from online AI support. The AI Tutor provides textbook grounded help for student questions using retrieval and answer generation. The AI component is optional. This ensures that modules, labels, info cards, Review Mode, and Quiz Mode continue to work even when network access or the AI service is unavailable.

This design keeps the project realistic for a graduation scope. It supports iterative content growth, stable classroom demos, and clear boundaries between interactive learning, review, assessment, and optional AI assistance.

4. Implementation Details

4.1. Introduction

The rapid advancement of virtual reality (VR) technologies has opened new opportunities in educational environments, particularly in fields that require strong visual and spatial understanding such as human anatomy. Traditional anatomy education methods are often limited in terms of interactivity and student engagement. Therefore, integrating immersive technologies like VR can significantly enhance the learning experience.

This project, titled “VR Anatomy: VR Based Educational Interactive Human Anatomy Training Platform”, aims to develop an interactive virtual reality application that enables users to explore human anatomy in a more engaging and intuitive way. The system allows users to interact with 3D anatomical models, examine structures closely, and access relevant information in real time.

The development process of the project is divided into two main phases: CENG 407 and CENG 408. In CENG 407, a basic working prototype was developed, focusing on core VR interactions and simple model visualization. In CENG 408, the project was expanded into a full-featured application by integrating multiple modules such as interactive learning modes, free exploration, quiz systems, and AI-assisted interaction.

This report focuses on the implementation details of the project in CENG 408, including the developed features, iteration process, system improvements, and technical components used throughout the development lifecycle.

4.2. System Overview

The VR Anatomy system is developed using Unity 6 as the main development platform and is designed to run on Meta Quest 2 virtual reality devices. The system provides an immersive and interactive learning environment where users can explore human anatomy through 3D models and intuitive VR interactions.

The application is built using the XR Interaction Toolkit, which enables natural user interactions such as grabbing, rotating, and inspecting anatomical structures. Users can navigate through different modes and interact with objects using VR controllers.

The system architecture consists of several core modules.

4.2.1. Learning Mode

In this mode, users are guided through anatomical structures in a structured learning flow. The system presents anatomical parts along with visual explanations and contextual information. Additionally, checkpoint-based mini questions are included at certain stages to reinforce understanding. This combination ensures both informative and interactive learning.

4.2.2. Free Explore Mode

This mode allows users to freely interact with anatomical models without guidance. Users can grab and examine objects, and optionally display the names of the selected structures. Unlike the learning mode, detailed information panels are hidden to encourage self-exploration.

4.2.3. Quiz System

The quiz module is designed to evaluate the user's knowledge. It includes categorized questions, a countdown system before the quiz starts, and interactive answer selection within the VR environment.

4.2.4. AI Interaction System

The system includes an AI-assisted interaction module that enables users to communicate with an intelligent assistant. This module integrates speech-to-text (STT) and text-to-speech (TTS) functionalities, allowing users to ask questions verbally and receive spoken responses. The AI backend is supported by a Retrieval-Augmented Generation (RAG) approach, providing context-aware answers based on anatomical data.

4.2.5. Scene and Navigation System

The application includes a structured navigation system that manages transitions between different scenes and modes. A centralized state management approach is used to maintain user context, such as selected modules and interaction modes.

4.3. Implemented Features

This section describes the core features implemented in the VR Anatomy system, focusing on interaction design, visual feedback mechanisms, and system functionality.

4.3.1. Object Interaction System

The system allows users to interact with anatomical models using VR controllers through the XR Interaction Toolkit. Objects in the scene are configured as interactable components, enabling users to grab, move, and inspect them in a natural way.

When a user selects an object, it can be freely rotated and examined from different angles, providing a more immersive understanding of the anatomical structure. This interaction model enhances spatial awareness and supports experiential learning.

4.3.2. Visual Filtering System(Context-Based Visibility)

A context-based visual filtering system is implemented to improve clarity during the Free Explore mode. When a specific anatomical unit is selected, only the models related to that unit remain fully visible with their original appearance.

All unrelated anatomical structures are visually de-emphasized by reducing their opacity, creating a semi-transparent effect. This allows users to focus on the relevant components without completely removing the surrounding context.

Unlike traditional highlight systems, no visual enhancement is applied when an object is selected or grabbed. Instead, the system emphasizes contextual relevance by distinguishing related and unrelated structures through transparency.

This approach maintains the visual integrity of anatomical models while providing a clear and intuitive focus mechanism for the user.

4.3.3. Free Explore Mode Implementation

The Free Explore mode allows users to interact with anatomical models without guided instructions, providing a self-directed learning experience.

Users can freely grab, rotate, and inspect anatomical structures in a virtual environment. To keep the interface minimal and avoid visual clutter, object names are not displayed by default. Instead, an optional UI panel is provided.

Users can toggle the visibility of the object name panel using a button on the VR controller. When activated, the system displays the name of the currently held object. This ensures that users can access information when needed, without interrupting the immersive experience.

This design balances usability and immersion by allowing on-demand information access rather than constant visual overlays.

4.3.4. Quiz System Implementation

The Quiz System is designed as an adaptive assessment mechanism to evaluate the user's knowledge and guide their learning process.

Users access the quiz from the main menu and select a specific anatomical category (e.g., Circulatory System or Musculoskeletal System). Before the quiz begins, an informational screen is displayed, explaining the rules such as time limits, question types, and the countdown process.

Each quiz session includes:

- Multiple-choice questions
- Matching-type questions
- A time constraint (e.g., 30 seconds per question)
- A countdown sequence before the quiz starts

Questions are internally categorized into different difficulty levels (easy, medium, hard). The system dynamically adjusts the difficulty based on the user's performance:

- If the user answers correctly, subsequent questions become more difficult
- If the user struggles, the system provides easier questions

This adaptive mechanism allows the system to evaluate the user more accurately according to their level.

At the end of the quiz, the system analyzes the user's performance and provides feedback based on topic-wise accuracy. Based on this analysis, users are redirected to the relevant learning modules where they have weaknesses.

This creates a feedback loop between assessment and learning, making the system both evaluative and instructive.

4.3.5. AI-Based Interactive Learning System

An AI-based interactive learning system is integrated into the application to enhance user engagement and provide personalized educational support.

The system utilizes a Retrieval-Augmented Generation (RAG) architecture combined with the Gemini API to generate accurate and context-aware responses. Anatomical knowledge is stored in a structured dataset, and relevant information is retrieved and provided to the AI model to ensure reliable answers.

Users can interact with the AI system through both text and voice-based interfaces:

- Text-based interaction: Users can type questions and receive generated responses
- Voice-based interaction: Speech-to-Text (STT) is used to convert user speech into text, and Text-to-Speech (TTS) is used to deliver responses audibly

This allows for a more natural and immersive communication experience within the VR environment.

The AI system is designed to answer questions related to anatomical structures, explain concepts, and support the learning process dynamically. By combining RAG with multimodal interaction (text + voice), the system provides both accessibility and flexibility for different user preferences.

4.4. Implementation Details

This section presents the development process of the VR Anatomy project, including versioning, iteration workflow, repository reference, and system outputs.

4.4.1. Versioning

The project was developed using an internal versioning approach to track the progression of system features. Instead of formal public releases, versioning was used conceptually to represent major development milestones.

The versioning structure reflects the internal development stages of the system:

- **v0.1** – Initial VR setup and basic interaction system
- **v0.2** – Early Learning Mode structure
- **v0.3** – Free Explore mode and visual filtering system
- **v0.4** – Quiz system with adaptive difficulty
- **v0.5** – AI integration (RAG + Gemini API + STT/TTS)
- **v1.0** – Final integrated system (in progress)

4.4.2. Iteration Log

The development process followed an iterative and non-linear approach. Different components of the system were developed in parallel and continuously refined.

Instead of a strictly sequential workflow, the team worked on multiple modules simultaneously, including interaction systems, AI integration, and user interface design.

The iterations can be conceptually summarized as follows:

- **Iteration 1:** Initial VR setup and core interaction system
- **Iteration 2:** Learning Mode development
- **Iteration 3:** Free Explore mode and context-based visual filtering
- **Iteration 4:** Quiz system with adaptive difficulty mechanism
- **Iteration 5:** AI-based system integration (RAG + Gemini API + voice interaction)
- **Iteration 6:** UI improvements, optimization, and ongoing refinements

This flexible development approach allowed continuous testing, feedback integration, and progressive improvement of the system.

4.4.3. Repository Reference

The project source code is maintained in a GitHub repository:

- <https://github.com/CankayaUniversity/ceng-407-408-2025-2026-VR-Anatomy-VR-Based-Educational-Interactive-Human-Anatomy-Training-Platform>

The repository includes:

- Unity project files
- AI backend components
- Version history and commit records
- Supporting resources and documentation

4.4.4. Screenshots

Screenshots are provided to demonstrate the implemented features and user interface of the VR Anatomy system. The images are organized according to the user flow within the application.

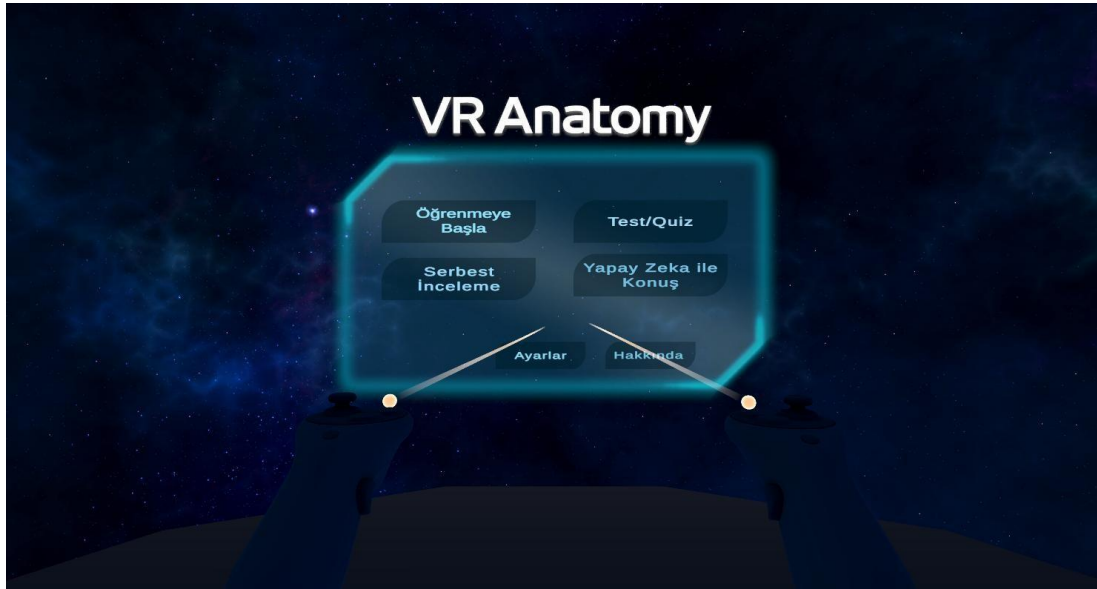


Figure 8: Main Menu Interface



Figure 9: Free Explore Mode with Visual Filtering



Figure 10: Free Explore Mode with Object Interaction and UI Panel

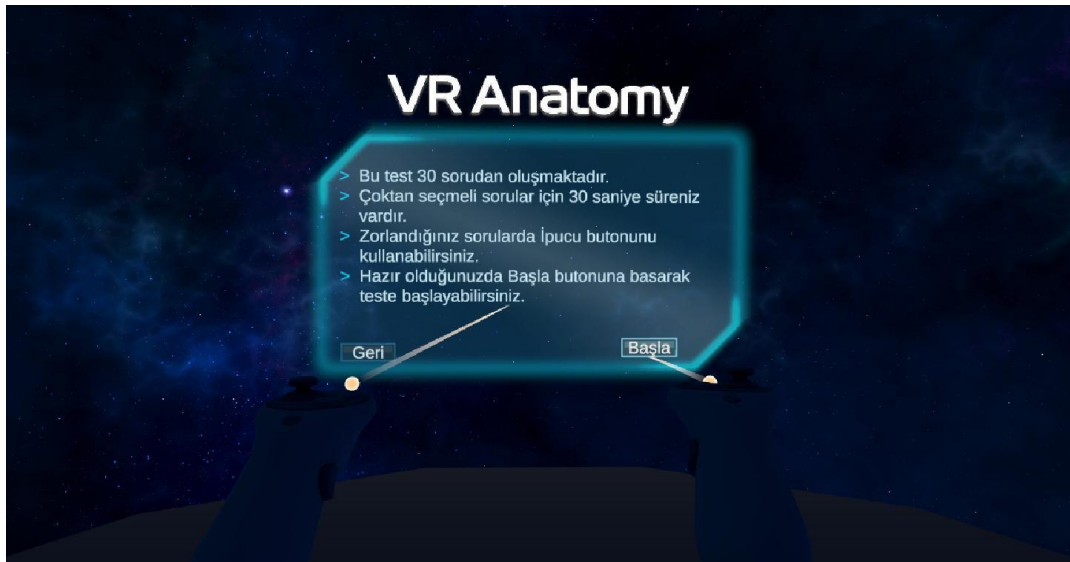


Figure 11: Quiz Information Screen

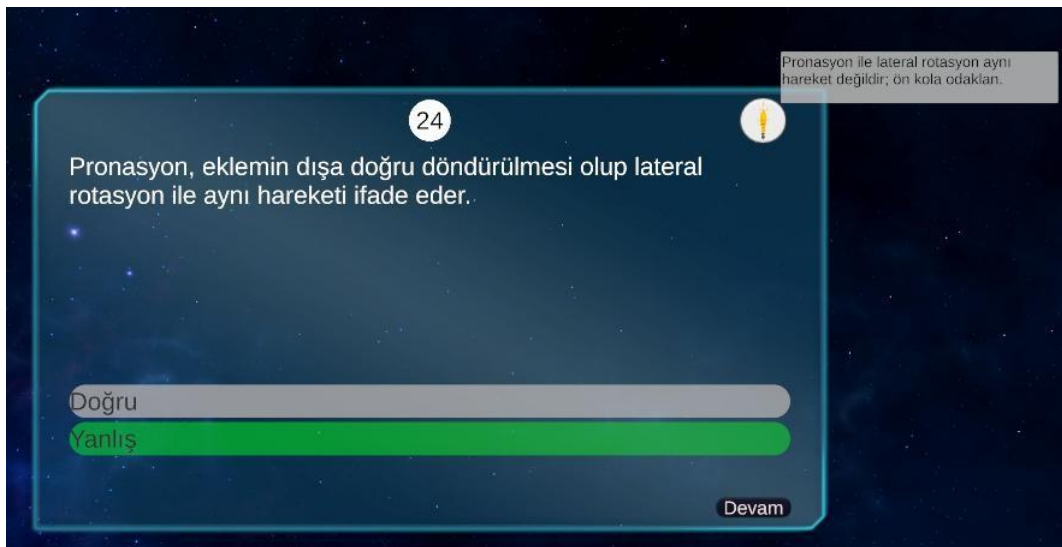


Figure 12: Quiz Mode Interface (True/False Question)

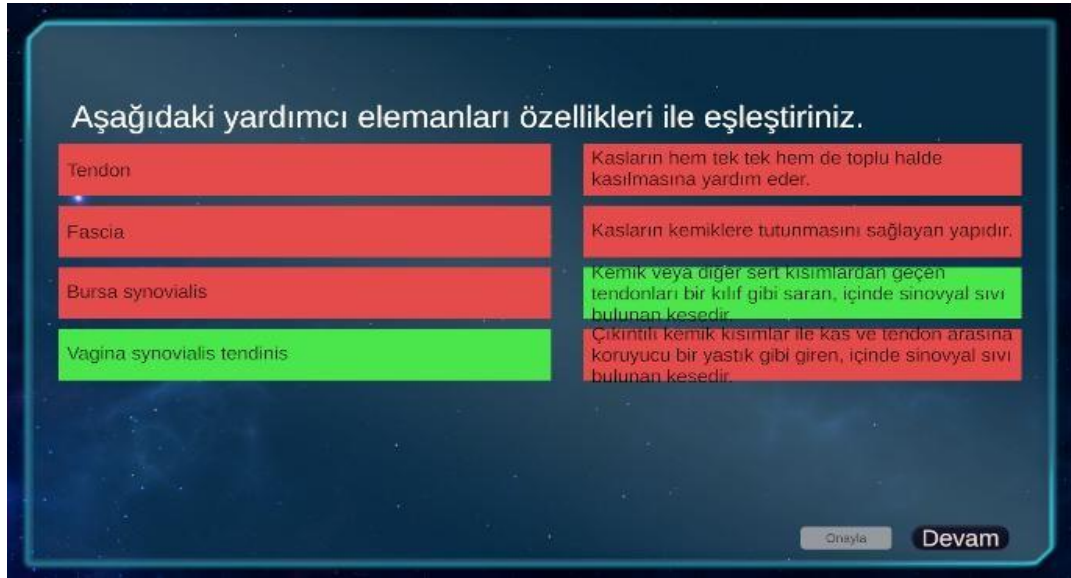


Figure 13: Quiz Mode Interface (Matching Question)

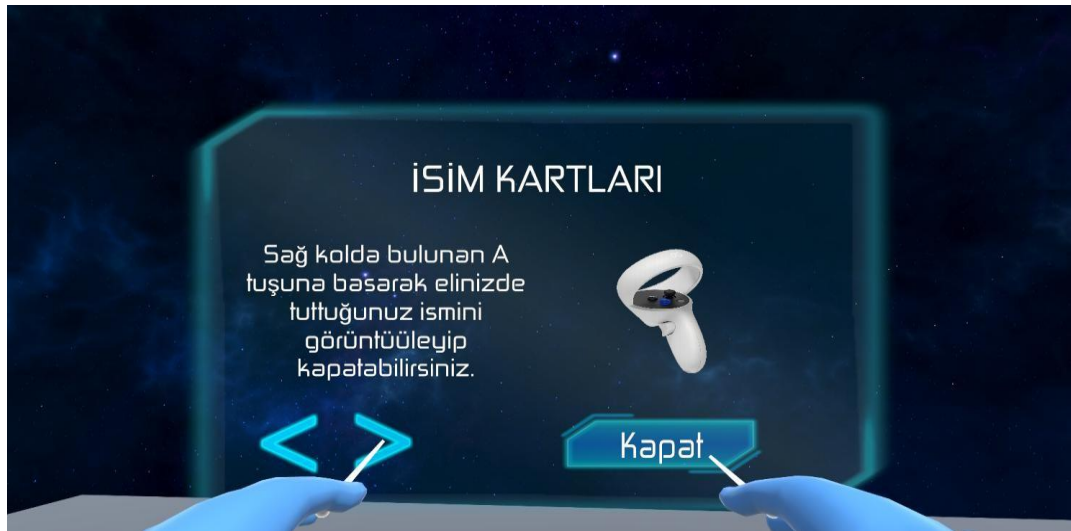


Figure 14: Instructional UI Panel (User Guidance System)



Figure 15: Learning Mode Interface



Figure 16: AI-Based Interaction Interface

5. Testing & Validation

5.1. Introduction

5.1.1. Version Control

Version No	Description of Changes	Date
1.0	First version of the test plan document	April 23, 2026

5.1.2. Overview

The current document describes the test plan, test design, and test cases related to the VR Anatomy project. Testing is performed to verify the proper functioning of the system in the virtual reality environment, especially the process of visualizing body organs, navigating through virtual space, interacting with objects, AI-supported chatbot, and providing learning opportunities.

5.1.3. Scope

This document covers the functional testing of major features of the VR Anatomy application, including the visualization of the anatomical model, selecting objects, navigating through scenes, menu interaction, chatbot responses, and quizzes. It will also cover pass/fail criteria and test cases for implemented modules.

5.1.4. Terminology

Acronym	Definition
VR	Virtual Reality
FE	Free Explore
QZ	Quiz Module
CB	Chatbot
SN	Scene Navigation
MI	Menu Interaction
TC	Test Case

5.2. FEATURES TO BE TESTED

This section lists and gives a brief description of all the major features to be tested. For each major feature there will be a Test Design Specification added at the end of this document.

5.2.1. Free Explore(FE)

This feature enables users to interact with anatomical structure using VR controllers. When the user selects a structure, the system highlights it and may display related information.

5.2.2. Quiz Module(QZ)

This feature provides quiz-based interaction to reinforce anatomical knowledge. Users can answer questions and receive feedback based on their selections.

5.2.3. Chatbot(CB)

This feature allows users to communicate with an AI-based chatbot about anatomy using voice input and output. The chatbot is supported by an interactive avatar that provides visual feedback during interaction.

5.2.4. Scene Navigation(SN)

This feature enables movement between scenes or sections of the application. It ensures that transitions between modules are correctly handled.

5.2.5. Menu Interaction(MI)

This feature includes the main menu and in-application user interface interactions. It allows the user to start the experience, navigate between options, and access available learning modules.

5.3. FEATURES NOT TO BE TESTED

- The “Learning Mode” feature is currently under development and is not included in the testing process at this stage.

5.4. ITEM PASS/FAIL CRITERIA

When the real output of the software is found to be the same as the predicted output mentioned in the test case, then the test case is said to have passed. When there is a difference between the expected and actual behavior of the software system, then the test case fails.

5.4.1. Exit Criteria

Testing is considered successful when all planned test cases are executed, at least 90% of the test cases pass, and no critical functional errors remain in the application. High-priority test cases must pass before final submission.

5.5. REFERENCES

- [1] VR Anatomy Literature Review, Version 1.0, 2026
- [2] VR Anatomy Product Vision & Scope Document, Version 1.0, 2026
- [3] VR Anatomy System Architecture & Design Decisions Document, Version 1.0, 2026
- [4] VR Anatomy Implementation Details Document, Version 1.0, 2026
- [5] VR Anatomy Project Report (CENG 407), 2026

5.6. TEST DESIGN SPECIFICATIONS

5.6.1. Free Explore(FE)

5.6.1.1. Subfeatures to be tested

5.6.1.1.1. Highlighting selected region(FE.HL)

After the user selects a specific anatomical region, this subfeature allows the system to display the related anatomical view. The surrounding structures gradually become transparent, while the selected region is visually emphasized for detailed examination.

5.6.1.1.2. Displaying Sturcture information(FE.DI)

This subfeature shows the corresponding name after anatomical structure selection.

5.6.1.1.3. Info Screen(FE.IS)

This subfeature displays an information screen that explains VR controller usage before the user begins exploration.

5.6.1.2. Requirements

R1: The system shall display a focused anatomical view when a region is selected and visually emphasize the selected region while surrounding structures become transparent.

R2: The system shall display the name or related information of the selected anatomical structure.

R3: The system shall display an instructional information screen explaining VR controller usage before the exploration phase begins.

5.6.1.3. Test Cases

Here list all the related test cases for this feature

TC ID	Requirements	Priority	Scenario Description
FE.HL.01	R1	H	Select an anatomical region and verify that the system displays the focused view with surrounding structures becoming transparent
FE.HL.02	R1	M	Select different anatomical regions and verify that the correct region is emphasized each time
FE.DI.01	R2	H	Select an anatomical structure and verify that its name is displayed
FE.IS.01	R3	H	Enter Free Explore mode and verify that the instruction screen is displayed before interaction
FE.IS.02	R3	M	Proceed from the instruction screen and verify that exploration starts correctly
FE.IS.03	R3	L	Skip the instruction screen and verify system behavior

5.6.2. Quiz Module(QZ)

5.6.2.1. Subfeatures to be tested

5.6.2.1.1. Quiz Introduction Screen (QZ.IN)

This subfeature displays an informational card before the quiz begins, providing details about the quiz flow and instructions to the user.

5.6.2.1.2. Countdown Mechanism (QZ.CD)

This subfeature initiates a countdown before the quiz starts, allowing the user to prepare before answering questions.

5.6.2.1.3. Answer Evaluation and Feedback (QZ.AF)

This subfeature evaluates the user's answer and provides immediate visual feedback. Correct answers are highlighted in green, while incorrect answers are highlighted in red, followed by an informational card explaining the correct answer.

5.6.2.1.4. Time Management (QZ.TM)

This subfeature tracks the time allocated for each question. If the user fails to answer within the given time, the question is marked as unanswered.

5.6.2.1.5. Hint System (QZ.HS)

This subfeature provides a hint option if the user does not respond within a certain time, assisting the user in answering the question.

5.6.2.1.6. Result Analysis (QZ.RA)

This subfeature generates an analysis of the user's performance after the quiz is completed, based on their answers.

5.6.2.1.7. Adaptive Question Difficulty (QZ.AD)

This subfeature dynamically adjusts the difficulty of questions. Initially, medium-level questions from different topics are presented. Based on the user's performance, subsequent questions are selected as easy or hard, depending on correctness.

5.6.2.2. Requirements

R4: The system shall display an informational screen before the quiz begins, providing instructions and details about the quiz.

R5: The system shall initiate a countdown before the quiz starts.

R6: The system shall evaluate the user's answer and provide immediate visual feedback indicating whether the answer is correct or incorrect.

R7: The system shall display an informational explanation for incorrect answers.

R8: The system shall limit the time allocated for each question.

R9: The system shall mark a question as unanswered if the user fails to respond within the given time.

R10: The system shall provide a hint option if the user does not answer within a specified time.

R11: The system shall generate a performance analysis based on the user's answers after the quiz is completed.

R12: The system shall present initial questions at medium difficulty from different topics.

R13: The system shall adjust subsequent question difficulty based on the user's previous answers.

5.6.2.3. Test Cases

Here list all the related test cases for this feature

TC ID	Requirements	Priority	Scenario Description
QZ.IN.01	R4	H	Start the quiz and verify that an informational screen is displayed before the quiz begins
QZ.CD.01	R5	H	Proceed from the introduction screen and verify that a countdown is displayed before the first question appears
QZ.AF.01	R6	H	Answer a question correctly and verify that the selected option is highlighted in green
QZ.AF.02	R6, R7	H	Answer a question incorrectly and verify that the selected option is highlighted in red and an explanation card is displayed
QZ.AF.03	R7	M	Verify that the explanation card shown after an incorrect answer matches the correct answer and related information
QZ.TM.01	R8	H	Start a question and verify that the system applies a time limit for answering
QZ.TM.02	R9	H	Do not answer a question until the time expires and verify that the question is marked as unanswered
QZ.HS.01	R10	H	Wait for a certain period without answering and verify that the hint button appears
QZ.RA.01	R11	H	Complete the quiz and verify that a performance analysis screen is displayed based on the user's answers
QZ.RA.02	R11	M	Complete the quiz with different answer patterns and verify that the analysis result changes accordingly

QZ.AD.01	R12	H	Start the quiz and verify that the initial questions are medium-level questions selected from different topics
QZ.AD.02	R13	H	Answer topic questions correctly and verify that the following questions become harder
QZ.AD.03	R13	H	Answer topic questions incorrectly and verify that the following questions become easier

5.6.3. Chatbot (CB)

5.6.3.1. Subfeatures to be tested

5.6.3.1.1. Instruction Screen (CB.IS)

This subfeature displays an information screen before the chatbot interaction begins.

5.6.3.1.2. Speech-to-Text Input (CB.ST)

This subfeature converts the user's spoken input into text when the user presses the talk button, allowing voice-based interaction with the chatbot.

5.6.3.1.3. Automatic Question Submission (CB.AS)

This subfeature automatically submits the user's question when the spoken input ends with the command "cevapla," enabling hands-free interaction.

5.6.3.1.4. Manual Question Submission (CB.MS)

This subfeature allows the user to manually send the recognized text as a question by pressing the ask button if automatic submission is not triggered.

5.6.3.1.5. AI-Based Response Generation (CB.AR)

This subfeature sends the user's question to the AI-based Gemini model in the background and retrieves the generated response.

5.6.3.1.6. Avatar-Supported Response Display (CB.AV)

This subfeature presents the chatbot response together with the avatar, providing a visual representation of the interaction.

5.6.3.1.7. Audio Response Output (CB.AO)

This subfeature enables the avatar to explain the chatbot's response through voice output when the user presses the listen button.

5.6.3.2. Requirements

R14: The system shall display an instructional screen explaining the chatbot interface and button functionalities before the interaction begins.

R15: The system shall convert the user's spoken input into text when the talk button is pressed.

R16: The system shall automatically submit the user's question when the spoken input ends with the command "cevapla."

R17: The system shall allow the user to manually submit the recognized text by pressing the ask button.

R18: The system shall send the user's question to a Gemini-based AI model and retrieve a generated response.

R19: The system shall display the chatbot's response together with the avatar.

R20: The system shall provide an audio output of the chatbot's response when the listen button is pressed.

5.6.3.3. Test Cases

Here list all the related test cases for this feature

TC ID	Requirements	Priority	Scenario Description
CB.IS.01	R14	H	Enter the chatbot module and verify that an instructional screen explaining the interface and button functions is displayed before interaction begins

CB.ST.01	R15	H	Press the talk button, speak a question, and verify that the spoken input is converted into text
CB.ST.02	R15	M	Speak different anatomy-related questions and verify that the recognized text is displayed correctly
CB.AS.01	R16	H	Press the talk button, speak a question ending with “cevapla,” and verify that the system automatically submits the question
CB.AS.02	R16	M	Speak a question without ending it with “cevapla” and verify that the system does not automatically submit it
CB.MS.01	R17	H	Press the talk button, speak a question without saying “cevapla,” then press the ask button and verify that the question is submitted manually
CB.AR.01	R18	H	Submit an anatomy-related question and verify that the system sends it to the Gemini-based AI model and receives a response
CB.AR.02	R18	M	Submit multiple questions and verify that the system generates responses for each of them
CB.AV.01	R19	H	Submit a question and verify that the chatbot response is displayed together with the avatar
CB.AV.02	R19	M	Verify that the avatar remains visible during the response display
CB.AO.01	R20	H	Press the listen button after receiving a response and verify that the system provides audio output of the response
CB.AO.02	R20	H	Verify that the audio output matches the displayed chatbot response when the listen button is pressed

5.6.4. Scene Navigation (SN)

5.6.4.1. Subfeatures to be tested

5.6.4.1.1. Scene Transition from Menu (SN.MT)

This subfeature allows the user to navigate from the main menu to the related scene by selecting the corresponding option.

5.6.4.1.2. In-Scene Navigation (SN.IN)

This subfeature allows the user to move between different sections or scenes within the application through available navigation controls.

5.6.4.1.3. Return to Menu (SN.RM)

This subfeature allows the user to return to the main menu from an active scene using the defined input or navigation option

5.6.4.2. Requirements

R21: The system shall allow the user to navigate from the main menu to the corresponding scene by selecting the related option.

R22: The system shall allow the user to navigate between different sections or scenes within the application using the available navigation controls.

R23: The system shall allow the user to return to the main menu from an active scene using the defined input or navigation

5.6.4.3. Test Cases

Here list all the related test cases for this feature

TC ID	Requirements	Priority	Scenario Description
SN.MT.01	R21	H	Verify that selecting an option from the main menu opens the correct scene
SN.IN.01	R22	H	Verify that the user can navigate between scenes or sections using available controls
SN.RM.01	R23	H	Verify that the user can return to the main menu from an active scene
SN.RM.02	R23	M	Verify that the main menu is loaded correctly after returning from different active scenes

5.6.5. Menu Interaction (MI)

5.6.5.1. Subfeatures to be tested

5.6.5.1.1. Intro / Onboarding Screen (MI.IN)

This subfeature displays an introductory scene at the start of the application, which can be skipped by the user and is shown only once per session.

5.6.5.1.2. Menu Display (MI.MD)

This subfeature ensures that the main menu is displayed correctly and all menu elements are visible and accessible to the user.

5.6.5.1.3. Menu Option Selection (MI.OS)

This subfeature allows the user to select menu options using the VR controller and interact with the menu interface.

5.6.5.1.4. Button Functionality (MI.BF)

This subfeature ensures that each menu button performs its intended action when selected by the user.

5.6.5.1.5. Menu Flow Control (MI.FC)

This subfeature ensures that the user can navigate through the menu smoothly and access different parts of the application without errors.

5.6.5.2. Requirements

R24: The system shall display an introductory scene when the application is launched and allow the user to either watch or skip it.

R25: The system shall display the main menu correctly with all menu elements visible and accessible to the user.

R26: The system shall allow the user to select menu options using the VR controller.

R27: The system shall perform the intended action of each menu button when selected by the user.

R28: The system shall allow the user to navigate through the menu smoothly and access different parts of the application without errors.

5.6.5.3. Test Cases

Here list all the related test cases for this feature

TC ID	Requirements	Priority	Scenario Description
MI.IN.01	R24	H	Verify that the introductory scene is displayed when the application starts
MI.IN.02	R24	H	Verify that the user can skip the introductory scene and access the main menu
MI.MD.01	R25	H	Verify that the main menu is displayed correctly
MI.OS.01	R26	H	Verify that the user can select menu options using the VR controller
MI.BF.01	R27	H	Verify that each menu button performs its intended action
MI.FC.01	R28	H	Verify that the user can navigate through the menu and access different parts of the application without errors

5.7. DETAILED TEST CASES

5.7.1. FE.HL.01

TC_ID	FE.HL.01
Purpose	Verify that the selected anatomical region is displayed in a focused view and emphasized visually.
Requirements	R1
Priority	High.
Estimated Time Needed	3 Minutes
Dependency	The Free Explore scene and anatomical models must be loaded correctly.
Setup	The application is running, and the user is in Free Explore mode.
Procedure	[A01] Enter Free Explore mode. [A02] Select an anatomical region using the VR controller. [V01] Observe the displayed anatomical view. [V02] Check whether surrounding structures become transparent. [V03] Verify that the selected region is visually emphasized.
Cleanup	Return to the initial state or deselect the region.

5.7.2 FE.HL.02

TC_ID	FE.HL.02
Purpose	Verify that different anatomical regions are correctly emphasized when selected.
Requirements	R1
Priority	Medium.
Estimated Time Needed	4 Minutes
Dependency	FE.HL.01 should pass.
Setup	User is in Free Explore mode.
Procedure	[A01] Select one anatomical region. [A02] Observe the highlighted area. [A03] Select a different anatomical region. [V01] Verify that the newly selected region is emphasized. [V02] Verify that the previous selection is no longer emphasized.
Cleanup	Reset the selection.

5.7.3 FE.DI.01

TC_ID	FE.DI.01
Purpose	Verify that the name of the selected anatomical structure is displayed.
Requirements	R2
Priority	High.
Estimated Time Needed	2 Minutes
Dependency	The anatomical selection system must be functional.
Setup	User is in Free Explore mode.
Procedure	[A01] Select an anatomical structure.
	[V01] Observe the displayed information.
	[V02] Verify that the name of the selected structure appears on the screen.
Cleanup	Deselect the structure.

5.7.4 FE.IS.01

TC_ID	FE.IS.01
Purpose	Verify that the instruction screen is displayed before interaction begins.
Requirements	R3
Priority	High.
Estimated Time Needed	2 Minutes
Dependency	Application must be able to enter Free Explore mode.
Setup	Application is at the main menu.
Procedure	[A01] Select Free Explore mode from the menu.
	[V01] Observe whether the instruction screen appears.
	[V02] Verify that controller usage information is displayed.
Cleanup	Proceed to the next screen.

5.7.5 FE.IS.02

TC_ID	FE.IS.02
Purpose	Verify that the system correctly transitions to exploration after the instruction screen.
Requirements	R3
Priority	Medium.
Estimated Time Needed	2 Minutes

Dependency	FE.IS.01 should pass.
Setup	Instruction screen is displayed.
Procedure	[A01] Continue from the instruction screen.
	[V01] Observe whether the exploration mode starts.
	[V02] Verify that the system becomes interactive.
Cleanup	Return to menu if needed.

5.7.6 FE.IS.03

TC_ID	FE.IS.03
Purpose	Verify system behavior when the instruction screen is skipped.
Requirements	R3
Priority	Low.
Estimated Time Needed	2 Minutes
Dependency	Instruction screen must support skipping.
Setup	Instruction screen is displayed.
Procedure	[A01] Skip the instruction screen.
	[V01] Observe system behavior.
	[V02] Verify that the exploration mode starts correctly.
Cleanup	Return to menu.

5.7.7 QZ.IN.01

TC_ID	QZ.IN.01
Purpose	Verify that an informational screen is displayed before the quiz begins.
Requirements	R4
Priority	High.
Estimated Time Needed	2 Minutes
Dependency	The quiz module should be accessible from the menu.
Setup	The application is running and the user is on the menu screen.
Procedure	[A01] Open the quiz module from the menu.
	[V01] Observe whether an informational screen appears before the quiz starts.
	[V02] Verify that the screen provides quiz-related instructions and information.
Cleanup	Proceed to the next screen.

5.7.8 QZ.CD.01

TC_ID	QZ.CD.01
Purpose	Verify that a countdown is displayed before the first question appears.
Requirements	R5
Priority	High.
Estimated Time Needed	1 Minutes
Dependency	QZ.IN.01 should pass.
Setup	The informational screen is displayed.
Procedure	[A01] Continue from the informational screen.
	[V01] Observe whether a countdown starts before the first question appears.
	[V02] Verify that the first question is displayed immediately after the countdown ends.
Cleanup	Continue the quiz.

5.7.9 QZ.AF.01

TC_ID	QZ.AF.01
Purpose	Verify that the selected option is highlighted in green when the user answers correctly.
Requirements	R6
Priority	High.
Estimated Time Needed	3 Minutes
Dependency	The quiz question and answer options should be displayed correctly.
Setup	A quiz question is active on the screen.
Procedure	[A01] Read the displayed quiz question.
	[A02] Select the correct answer option.
	[V01] Observe the selected option after answering.
	[V02] Verify that the selected option is highlighted in green.
Cleanup	Proceed to the next question.

5.7.10 QZ.AF.02

TC_ID	QZ.AF.02
Purpose	Verify that the selected option is highlighted in red and an explanation card is displayed when the user answers incorrectly.
Requirements	R6, R7
Priority	High.
Estimated Time Needed	3 Minutes
Dependency	The quiz question and answer options should be displayed correctly.
Setup	A quiz question is active on the screen.
Procedure	[A01] Read the displayed quiz question.
	[A02] Select an incorrect answer option.
	[V01] Observe the selected option after answering.
	[V02] Verify that the selected option is highlighted in red.
	[V03] Verify that an explanation card is displayed after the incorrect answer.
Cleanup	Close the explanation card and continue the quiz.

5.7.11 QZ.AF.03

TC_ID	QZ.AF.03
Purpose	Verify that the explanation card shown after an incorrect answer matches the correct answer and related information.
Requirements	R7
Priority	Medium.
Estimated Time Needed	3 Minutes
Dependency	QZ.AF.02 should pass.
Setup	An explanation card is displayed after an incorrect answer.
Procedure	[A01] Answer a question incorrectly.
	[V01] Observe the explanation card displayed on the screen.
	[V02] Verify that the explanation card includes the correct answer.
	[V03] Verify that the explanation content is related to the question.
Cleanup	Close the explanation card.

5.7.12 QZ.TM.01

TC_ID	QZ.TM.01
Purpose	Verify that the system applies a time limit for answering each question.
Requirements	R8
Priority	High.
Estimated Time Needed	1 Minutes
Dependency	A quiz question should be active on the screen.
Setup	The quiz has started and a question is displayed.
Procedure	[A01] Start a quiz question.
	[V01] Observe the question screen.
	[V02] Verify that the system displays or applies a time limit for answering.
Cleanup	Proceed to the next question or return to the quiz flow.

5.7.13 QZ.TM.02

TC_ID	QZ.TM.02
Purpose	Verify that a question is marked as unanswered when the time expires.
Requirements	R9
Priority	High.
Estimated Time Needed	1 Minutes
Dependency	QZ.TM.01 should pass.
Setup	A timed quiz question is displayed.
Procedure	[A01] Start the question and do not select any answer.
	[A02] Wait until the time expires.
	[V01] Observe the result after the timer ends.
	[V02] Verify that the question is marked as unanswered.
Cleanup	Proceed to the next question.

5.7.14 QZ.HS.01

TC_ID	QZ.HS.01
Purpose	Verify that the hint button appears after a certain period without answering.
Requirements	R10
Priority	High.
Estimated Time Needed	1 Minutes
Dependency	A quiz question should remain unanswered for a certain duration.
Setup	A quiz question is displayed on the screen.
Procedure	[A01] Start a quiz question.
	[A02] Do not answer the question for a certain period.
	[V01] Observe the screen while waiting.
	[V02] Verify that the hint button appears after the required waiting time.
Cleanup	Continue the quiz.

5.7.15 QZ.RA.01

TC_ID	QZ.RA.01
Purpose	Verify that a performance analysis screen is displayed based on the user's answers after the quiz is completed.
Requirements	R11
Priority	High.
Estimated Time Needed	5 Minutes
Dependency	The quiz should be completed.
Setup	The user is at the final stage of the quiz.
Procedure	[A01] Complete all quiz questions.
	[V01] Observe the screen shown after the quiz ends.
	[V02] Verify that a performance analysis screen is displayed.
	[V03] Verify that the analysis is based on the user's answers.
Cleanup	Return to the menu.

5.7.16 QZ.RA.02

TC_ID	QZ.RA.02
Purpose	Verify that the analysis result changes according to different answer patterns.
Requirements	R11
Priority	Medium.
Estimated Time Needed	5 Minutes
Dependency	QZ.RA.01 should pass.
Setup	The quiz can be repeated with different answer patterns.
Procedure	[A01] Complete the quiz with mostly correct answers.
	[V01] Observe the analysis result.
	[A02] Restart the quiz.
	[A03] Complete the quiz with mostly incorrect or unanswered questions.
	[V02] Observe the new analysis result.
	[V03] Verify that the analysis result changes according to the user's performance.
Cleanup	Return to the menu.

5.7.17 QZ.AD.01

TC_ID	QZ.AD.01
Purpose	Verify that the initial questions are medium-level questions selected from different topics.
Requirements	R12
Priority	High.
Estimated Time Needed	3 Minutes
Dependency	The adaptive quiz system should be active.
Setup	The quiz has not started yet.
Procedure	[A01] Start the quiz.
	[V01] Observe the first questions presented by the system.
	[V02] Verify that the initial questions are medium-level.
	[V03] Verify that these questions are selected from different topics.
Cleanup	Continue the quiz.

5.7.18 QZ.AD.02

TC_ID	QZ.AD.02
Purpose	Verify that the following questions become harder when the user answers correctly.
Requirements	R13
Priority	High.
Estimated Time Needed	3 Minutes
Dependency	QZ.AD.01 should pass.
Setup	The quiz is running with adaptive difficulty enabled.
Procedure	[A01] Start the quiz.
	[A02] Answer the initial topic questions correctly.
	[V01] Observe the difficulty level of the following questions.
	[V02] Verify that the next questions become harder after correct answers.
Cleanup	Continue or finish the quiz.

5.7.19 QZ.AD.03

TC_ID	QZ.AD.03
Purpose	Verify that the following questions become easier when the user answers incorrectly.
Requirements	R13
Priority	High.
Estimated Time Needed	3 Minutes
Dependency	QZ.AD.01 should pass.
Setup	The quiz is running with adaptive difficulty enabled.
Procedure	[A01] Start the quiz.
	[A02] Answer the initial topic questions incorrectly.
	[V01] Observe the difficulty level of the following questions.
	[V02] Verify that the next questions become easier after incorrect answers.
Cleanup	Continue or finish the quiz.

5.7.20 CB.IS.01

TC_ID	CB.IS.01
Purpose	Verify that an instructional screen is displayed before chatbot interaction begins.
Requirements	R14
Priority	High.
Estimated Time Needed	2 Minutes
Dependency	The chatbot module should be accessible.
Setup	The application is running and the user is at the main menu.
Procedure	[A01] Enter the chatbot module.
	[V01] Observe whether the instructional screen appears.
	[V02] Verify that the screen includes explanations about buttons and interaction.
Cleanup	Proceed to the chatbot interface.

5.7.21 CB.ST.01

TC_ID	CB.ST.01
Purpose	Verify that spoken input is converted into text using speech-to-text.
Requirements	R15
Priority	High.
Estimated Time Needed	3 Minutes
Dependency	CB.IS.01 should pass.
Setup	The chatbot interface is open.
Procedure	[A01] Press the talk button.
	[A02] Speak a question clearly.
	[V01] Observe the recognized text.
	[V02] Verify that the spoken input is converted into text correctly.
Cleanup	Clear input field if necessary.

5.7.22 CB.ST.02

TC_ID	CB.ST.02
Purpose	Verify that different spoken inputs are correctly converted into text.
Requirements	R15
Priority	Medium.
Estimated Time Needed	3 Minutes
Dependency	CB.ST.01 should pass.
Setup	The chatbot interface is open and the talk button is functional.
Procedure	[A01] Press the talk button.
	[A02] Speak an anatomy-related question.
	[V01] Observe the recognized text.
	[A03] Repeat the process with a different question.
	[V02] Verify that the new spoken input is correctly converted into text.
	[V03] Compare the spoken input with the displayed text for accuracy.
Cleanup	Clear the input field if necessary.

5.7.23 CB.AS.01

TC_ID	CB.AS.01
Purpose	Verify that the system automatically submits the question when “cevapla” is used.
Requirements	R16
Priority	High.
Estimated Time Needed	3 Minutes
Dependency	CB.ST.01 should pass.
Setup	The chatbot interface is active.
Procedure	[A01] Press the talk button.
	[A02] Speak a question ending with “cevapla.”
	[V01] Observe whether the system automatically submits the question.
	[V02] Verify that the response is generated without pressing the ask button.
Cleanup	Wait for response completion.

5.7.24 CB.AS.02

TC_ID	CB.AS.02
Purpose	Verify that the system does not automatically submit the question when the “cevapla” command is not used.
Requirements	R16
Priority	Medium.
Estimated Time Needed	3 Minutes
Dependency	CB.ST.01 should pass.
Setup	The chatbot interface is open and the talk button is functional.
Procedure	[A01] Press the talk button.
	[A02] Speak a question without ending it with “cevapla.”
	[V01] Observe whether the recognized text appears.
	[V02] Verify that the system does not automatically submit the question.
	[V03] Verify that the user can still manually submit the question using the ask button.
Cleanup	Clear the input field if necessary.

5.7.25 CB.MS.01

TC_ID	CB.MS.01
Purpose	Verify that the user can manually submit a question.
Requirements	R17
Priority	High.
Estimated Time Needed	3 Minutes
Dependency	CB.ST.01 should pass.
Setup	The chatbot interface is open.
Procedure	[A01] Press the talk button.
	[A02] Speak a question without saying “cevapla.”
	[A03] Press the ask button.
	[V01] Observe whether the question is submitted.
	[V02] Verify that the system processes the question correctly.
Cleanup	Wait for response.

5.7.26 CB.AR.01

TC_ID	CB.AR.01
Purpose	Verify that the system retrieves a response from the Gemini-based AI model.
Requirements	R18
Priority	High.
Estimated Time Needed	4 Minutes
Dependency	CB.MS.01 or CB.AS.01 should pass.
Setup	The chatbot interface is active and connected.
Procedure	[A01] Submit an anatomy-related question.
	[V01] Observe whether a response is generated.
	[V02] Verify that the system retrieves the response from the AI model.
	[V03] Verify that the response is relevant to the question.
Cleanup	None.

5.7.27 CB.AR.02

TC_ID	CB.AR.02
Purpose	Verify that the system generates responses for multiple submitted questions.
Requirements	R18
Priority	Medium.
Estimated Time Needed	3 Minutes
Dependency	CB.AR.01 should pass.
Setup	The chatbot interface is active and connected to the AI model.
Procedure	[A01] Submit an anatomy-related question.
	[V01] Observe the generated response.
	[A02] Submit a different question.
	[V02] Observe the new response.
	[A03] Submit additional questions consecutively.
	[V03] Verify that each question receives a corresponding response.
	[V04] Verify that the responses are updated correctly for each input.
Cleanup	None.

5.7.28 CB.AV.01

TC_ID	CB.AV.01
Purpose	Verify that the chatbot response is displayed together with the avatar.
Requirements	R19
Priority	High.
Estimated Time Needed	2 Minutes
Dependency	CB.AR.01 should pass.
Setup	A chatbot response is available.
Procedure	[A01] Submit a question.
	[V01] Observe the chatbot response.
	[V02] Verify that the avatar is displayed along with the response.
	[V03] Verify that the avatar remains visible during interaction.
Cleanup	None.

5.7.29 CB.AV.02

TC_ID	CB.AV.02
Purpose	Verify that the avatar remains visible during the response display.
Requirements	R19
Priority	Medium.
Estimated Time Needed	2 Minutes
Dependency	CB.AV.01 should pass.
Setup	The chatbot interface is open and a response is available.
Procedure	[A01] Submit a question to the chatbot.
	[V01] Observe the response area during the system response.
	[V02] Verify that the avatar remains visible while the response is displayed.
	[V03] Verify that the avatar is not hidden or removed during the interaction.
Cleanup	None.

5.7.30 CB.AO.01

TC_ID	CB.AO.01
Purpose	Verify that the system provides audio output of the chatbot response.
Requirements	R20
Priority	High.
Estimated Time Needed	3 Minutes
Dependency	CB.AR.01 should pass.
Setup	A chatbot response is displayed.
Procedure	[A01] Press the listen button.
	[V01] Observe whether audio output starts.
	[V02] Verify that the response is spoken clearly.
Cleanup	Stop audio playback if needed.

5.7.31 CB.AO.02

TC_ID	CB.AO.02
Purpose	Verify that the audio output matches the displayed chatbot response.
Requirements	R20
Priority	High.
Estimated Time Needed	3 Minutes
Dependency	CB.AO.01 should pass.
Setup	A chatbot response is available.
Procedure	[A01] Press the listen button.
	[V01] Listen to the audio output.
	[V02] Compare the spoken output with the displayed text.
	[V03] Verify that both outputs match.
Cleanup	None.

5.7.32 SN.MT.01

TC_ID	SN.MT.01
Purpose	Verify that selecting an option from the main menu opens the correct scene.
Requirements	R21
Priority	High.
Estimated Time Needed	2 Minutes
Dependency	Menu Interaction test cases should pass.
Setup	The application is running and the main menu is displayed.
Procedure	[A01] Open the main menu.
	[A02] Select a menu option using the VR controller.
	[V01] Observe whether the selected scene is loaded.
	[V02] Verify that the loaded scene corresponds to the selected menu option.
	[V03] Verify that the scene is displayed without missing elements or errors.
Cleanup	Return to the main menu.

5.7.33 SN.IN.01

TC_ID	SN.IN.01
Purpose	Verify that the user can navigate between scenes or sections using available controls.
Requirements	R22
Priority	High.
Estimated Time Needed	3 Minutes
Dependency	SN.MT.01 should pass.
Setup	The user is inside an active scene with navigation controls available.
Procedure	[A01] Start from an active scene.
	[A02] Use the available navigation control (button/interaction).
	[A03] Trigger a transition to another section or scene.
	[V01] Observe whether the new scene or section is loaded.

	[V02] Verify that the navigation control responds correctly.
	[V03] Verify that the system remains interactive after navigation.
Cleanup	Return to the main menu.

5.7.34 SN.RM.01

TC_ID	SN.RM.01
Purpose	Verify that the user can return to the main menu from an active scene.
Requirements	R23
Priority	High.
Estimated Time Needed	2 Minutes
Dependency	SN.MT.01 should pass.
Setup	The user is inside an active scene.
Procedure	[A01] While in an active scene, press the return-to-menu input or button.
	[V01] Observe whether the main menu is loaded.
	[V02] Verify that the application returns to the main menu without errors.
	[V03] Verify that the menu is fully functional after returning.
Cleanup	None.

5.7.35 SN.RM.02

TC_ID	SN.RM.02
Purpose	Verify that the main menu is loaded correctly after returning from different active scenes.
Requirements	R23
Priority	Medium.
Estimated Time Needed	3 Minutes
Dependency	SN.RM.01 should pass.
Setup	The application contains multiple scenes.
Procedure	[A01] Navigate to a scene from the main menu.
	[A02] Return to the main menu.
	[A03] Navigate to a different scene.
	[A04] Return to the main menu again.

	[V01] Observe whether the main menu loads correctly each time.
	[V02] Verify that no visual or functional errors occur.
	[V03] Verify that the menu remains usable after multiple returns.
Cleanup	Return to the main menu or close the application.

5.7.36 MI.IN.01

TC_ID	MI.IN.01
Purpose	Verify that the introductory scene is displayed when the application starts.
Requirements	R24
Priority	High.
Estimated Time Needed	2 Minutes
Dependency	The application should start successfully.
Setup	The application is closed before the test begins.
Procedure	[A01] Launch the application.
	[V01] Observe whether the introductory scene appears at the start of the application.
	[V02] Verify that the introductory content is displayed correctly.
Cleanup	Close the application or proceed to the next screen.

5.7.37 MI.IN.02

TC_ID	MI.IN.02
Purpose	Verify that the user can skip the introductory scene and access the main menu.
Requirements	R24
Priority	High.
Estimated Time Needed	2 Minutes
Dependency	MI.IN.01 should pass.
Setup	The application has just been launched and the introductory scene is displayed.
Procedure	[A01] Launch the application.
	[A02] Wait until the introductory scene appears.

	[A03] Select the skip option.
	[V01] Observe whether the introductory scene is closed.
	[V02] Verify that the main menu is displayed correctly after skipping.
Cleanup	Return to the initial menu state.

5.7.38 MI.MD.01

TC_ID	MI.MD.01
Purpose	Verify that the main menu is displayed correctly.
Requirements	R25
Priority	High.
Estimated Time Needed	2 Minutes
Dependency	The application should be running correctly.
Setup	The user is on the main menu.
Procedure	[A01] Open the main menu.
	[V01] Observe the menu layout.
	[V02] Verify that all menu elements are visible.
	[V03] Verify that the menu is readable and accessible in the VR environment.
Cleanup	None.

5.7.39 MI.OS.01

TC_ID	MI.OS.01
Purpose	Verify that the user can select menu options using the VR controller.
Requirements	R26
Priority	High.
Estimated Time Needed	2 Minutes
Dependency	MI.MD.01 should pass.
Setup	The main menu is displayed and the VR controller is active.
Procedure	[A01] Point to a menu option using the VR controller.
	[A02] Press the selection input on the controller.

	[V01] Observe whether the selected menu option responds to the input.
	[V02] Verify that the system accepts the selection correctly.
Cleanup	Return to the main menu if another scene is opened.

5.7.40 MI.BF.01

TC_ID	MI.BF.01
Purpose	Verify that each menu button performs its intended action.
Requirements	R27
Priority	High.
Estimated Time Needed	4 Minutes
Dependency	MI.OS.01 should pass.
Setup	The user is on the main menu and all menu buttons are accessible.
Procedure	[A01] Select the first menu button.
	[V01] Verify that the corresponding function is executed correctly.
	[A02] Return to the main menu if necessary.
	[A03] Select another menu button.
	[V02] Verify that the second button performs its intended action.
	[A04] Repeat the same process for the remaining menu buttons.
Cleanup	Return to the main menu.

5.7.41 MI.FC.01

TC_ID	MI.FC.01
Purpose	Verify that the user can navigate through the menu and access different parts of the application without errors.
Requirements	R28
Priority	High.
Estimated Time Needed	4 Minutes
Dependency	MI.BF.01 should pass.
Setup	The application is running and the main menu is displayed.
Procedure	[A01] Open the main menu.
	[A02] Select one menu option and navigate to the related section.
	[A03] Return to the main menu.
	[A04] Select another menu option.
	[V01] Observe whether the user can move through the menu flow without interruption.
	[V02] Verify that different parts of the application can be accessed without errors.
	[V03] Verify that returning to and using the menu again does not cause any issue.
Cleanup	Return to the main menu or close the application.

6. Conclusion & Future Work

6.1. Introduction

Anatomy education requires students to understand the human body not only as a collection of separate structures, but also as a three-dimensional system in which organs, bones, muscles, vessels, and other anatomical parts are connected to each other. However, traditional learning materials such as textbooks, two-dimensional diagrams, and classroom presentations may not always be sufficient for students to visualize these spatial relationships clearly. This can make anatomy difficult to understand, especially for students who are learning the subject for the first time.

Virtual Reality (VR) offers a strong opportunity to support anatomy education by allowing students to experience anatomical structures in an immersive and interactive environment. Instead of only reading descriptions or looking at static images, students can observe 3D models from different angles, interact with anatomical parts, and explore the body in a more active way. This type of learning environment can make abstract or complex anatomical concepts more concrete and easier to remember.

At the same time, artificial intelligence can support the learning process by providing explanations, answering student questions, and helping learners review topics when they need additional guidance. When combined with VR, AI can make the learning experience more flexible and supportive by allowing students to receive information during exploration rather than depending only on fixed lesson content.

The main purpose of this project is to develop VR Anatomy, a VR-based educational platform that combines interactive anatomy visualization with AI-supported learning. The project aims to create a learning environment where students can explore selected anatomical systems, receive guided explanations, ask questions, and evaluate their understanding through quizzes. In this way, the system is designed to support traditional anatomy education with a more visual, interactive, and student-centered approach.

6.2. Project Overview

VR Anatomy is a functional prototype developed as a graduation project to support anatomy learning in a virtual reality environment. The system was designed for students who need visual and interactive support while learning introductory anatomy topics. The primary target group of the project is high school students taking anatomy and physiology courses, and the application was tested in a real classroom context with this target group during the first experimental condition.

The current scope of the application focuses on selected topics from the musculoskeletal system and the circulatory system. These systems were selected because they include structures that benefit from three-dimensional visualization, such as bones, muscles, the heart, and vessels. Due to time, content, and implementation constraints, the project does not cover the entire human anatomy. Instead, it presents a focused prototype that demonstrates how VR and AI can be used together in anatomy education.

The system includes four main modules. The AI Guided Learning module provides a structured learning flow in which students follow guided anatomical explanations. The Free Explore module allows students to examine anatomical models independently. The AI Chat module enables students to ask anatomy-related questions through text or voice and receive AI-supported answers. The Quiz module allows students to check their understanding after the learning process.

The project also includes avatar-based guidance and voice interaction. In the learning environment, the avatar supports the student by presenting explanations and creating a more natural interaction experience. The system includes speech-to-text and text-to-speech features so that students can communicate with the AI assistant by speaking or typing. These features aim to make the platform more interactive and accessible for different learning preferences.

From a technical perspective, the VR application was developed in Unity using XR interaction components. The artificial intelligence features were implemented through a Python-based backend that provides retrieval-supported answers from anatomy-related educational content. The backend also supports speech processing features used in the AI Chat and learning modules.

The project was also designed with two experimental interaction conditions. Experiment 1 uses a neutral learning experience with no student name collection, formal language, and a formal avatar. This condition has been completed and tested with high school students. Experiment 2 is planned as a follow-up condition that will use name-based personalization, friendlier language, and a more casual avatar. Therefore, the current version of the report presents the completed system and the initial evaluation status based on Experiment 1.

6.3. System Design and Architecture

The VR Anatomy system was designed as a combination of a Unity-based virtual reality application and a Python-based artificial intelligence backend. The Unity application is responsible for the visual learning environment, 3D anatomical models, user interaction, avatar presentation, quiz flow, and scene management. The backend is responsible for processing AI-related requests, retrieving relevant anatomy content, generating answers, and supporting speech-based interaction features.

The general architecture follows a client-server structure. The VR application acts as the client and sends requests to the backend when the user asks a question, requests a simplified explanation, or uses voice-based interaction. The backend processes these requests and returns the generated response to the Unity application. The response can then be displayed as text, read aloud through text-to-speech, or presented through the avatar depending on the active module.

The system was divided into multiple scenes and modules in Unity to keep the project organized and easier to manage. Each module has a specific purpose in the learning process. The AI Guided Learning module provides guided instruction, the Free Explore module supports independent model inspection, the AI Chat module handles open-ended question answering, and the Quiz module supports knowledge assessment. This modular structure made it easier to develop, test, and improve each part of the application separately.

The application uses 3D anatomical models to represent selected structures from the

musculoskeletal and circulatory systems. In AI Guided Learning, the relevant structure can be highlighted while unrelated structures can be hidden or made transparent. This design helps students focus on the current topic without being visually overwhelmed by all anatomical parts at once. In free exploration, students can interact with models more independently and examine them from different angles.

For artificial intelligence support, the backend uses a retrieval-based approach. Instead of generating answers without context, the system first searches anatomy-related educational content and retrieves relevant information. Then, the AI response is generated based on this retrieved content. This design was preferred to make the answers more consistent with the learning material and to reduce the risk of irrelevant or unsupported explanations.

The system also includes speech-to-text and text-to-speech features. Speech-to-text allows students to ask questions by speaking, while text-to-speech allows the system to read answers aloud. These features were added to make the application more natural in a VR environment, where typing can be less comfortable than speaking. Voice interaction also supports the avatar-based learning experience by making the communication feel more conversational.

Overall, the architecture of VR Anatomy connects interactive VR learning with AI-supported educational assistance. The Unity side provides the immersive and visual learning experience, while the backend provides intelligent explanation and voice support. This separation allows the system to remain modular, maintainable, and expandable for future improvements.

6.3.1. Unity VR Application

The main application was developed in Unity because Unity provides strong support for virtual reality development, 3D scene management, physics-based interaction, and cross-platform deployment. The VR environment was designed to allow students to observe anatomical structures in an immersive space and interact with them through VR controllers.

The Unity side of the project includes the main menu, module selection screens, anatomical learning scenes, free exploration scenes, quiz screens, avatar presentation, and user interface panels. The application was organized into separate scenes so that each learning module could be managed independently. This structure helped reduce complexity during development and made it easier to test individual parts of the system.

The project uses XR interaction components to support VR controller input, ray-based selection, object interaction, and model inspection. In the learning scenes, students can use controller-based interaction to select buttons, navigate between panels, and interact with anatomical models. In free exploration scenes, students can examine selected anatomical structures more independently.

A major design goal of the Unity application was to reduce visual complexity during anatomy learning. For this reason, the system can highlight the currently explained anatomical structure while making unrelated structures transparent or less visually dominant. This helps students focus on one topic at a time and understand the location of each structure within the

larger anatomical system.

The Unity application also includes avatar-based presentation. The avatar is used to make the learning experience more natural and guided. Depending on the active experimental condition, the avatar can represent either a more formal learning style or a more casual and friendly learning style. This design supports the project's experimental comparison between neutral and personalized interaction styles.

Overall, the Unity VR application forms the visual and interactive core of the project. It provides the environment where students experience anatomy through 3D models, guided explanations, free inspection, quizzes, and avatar-supported interaction.

6.3.2. AI and RAG Backend

The artificial intelligence component of the project was implemented as a Python-based backend. This backend is responsible for receiving requests from the Unity application, processing anatomy-related questions, retrieving relevant educational content, and returning generated answers to the VR interface.

The backend uses a retrieval-augmented generation approach. In this approach, the system does not rely only on the language model's general knowledge. Instead, it first searches the prepared anatomy learning content and retrieves relevant pieces of information. The generated answer is then formed based on this retrieved content. This makes the AI responses more connected to the course material and more suitable for an educational application.

The anatomy content used by the backend was prepared from educational sources related to the selected learning scope. The content was divided into smaller text chunks so that the system could search and retrieve the most relevant parts when a student asks a question. This structure allows the AI assistant to provide answers that are more focused on the user's question.

The backend is used in more than one part of the application. In the AI Chat module, students can ask open-ended anatomy questions and receive AI-supported answers. In the AI Guided Learning flow, the backend can also support simplified explanations when students need easier wording or additional clarification. This allows the system to provide adaptive support instead of presenting only fixed explanations.

One of the main reasons for using a separate backend is flexibility. The AI system can be improved, updated, or deployed independently from the Unity application. This also makes it possible to change the AI model, update the anatomy content, improve prompts, or add new features without rebuilding the entire VR application.

Overall, the AI and RAG backend provides the intelligent learning support layer of VR Anatomy. It helps the application move beyond static educational content by allowing students to ask questions and receive context-aware explanations during the learning process.

6.3.3. Speech-to-Text and Text-to-Speech Integration

Speech interaction was added to make the VR learning experience more natural and easier to use. Since typing in a VR environment can be difficult and slow, the system supports both speech-to-text and text-to-speech features. These features allow students to communicate with the AI assistant by speaking and to listen to the generated answers.

The speech-to-text feature allows students to record their questions through the microphone. The recorded voice input is sent to the backend, converted into text, and then processed as a question for the AI assistant. This makes the AI Chat module more suitable for VR use because students do not have to rely only on keyboard input.

The text-to-speech feature converts generated answers into spoken audio. After the AI assistant produces an answer, the system can display the response as text and also read it aloud. This is especially useful in a VR learning environment because students can continue looking at the 3D model while listening to the explanation.

Voice interaction is also connected to the avatar-based presentation. When the avatar presents or reads explanations, the learning experience becomes more conversational and guided. This supports the goal of making the application feel less like a static information screen and more like an interactive educational assistant.

The integration of speech-to-text and text-to-speech also supports accessibility and different learning preferences. Some students may prefer reading the text, while others may understand better by listening. By supporting both formats, the system provides a more flexible learning experience.

Overall, speech integration improves the usability of VR Anatomy by reducing the need for manual text input and by making explanations easier to follow inside the VR environment. It also strengthens the connection between the AI assistant, the avatar, and the student.

6.4. Implemented Modules

This section describes the user-facing functions implemented in each module of the VR Anatomy application.

6.4.1. AI Guided Learning Module

The AI Guided Learning module is the guided learning section of the application. In this module, students select an anatomy unit and follow the content in a structured order. The system presents anatomical structures one by one so that students do not need to explore the whole model without guidance.

During the learning flow, the currently explained anatomical structure is visually emphasized. Other structures can be made transparent or less dominant to help students focus on the active topic. Information panels are used to show the explanation of the selected structure, and

the avatar can support the process by presenting the content through voice.

The module also includes review support. When students need additional clarification, they can return to related structures or request a simpler explanation. This makes the module useful not only for first-time learning but also for repeating difficult topics.

6.4.2. Free Explore Module

The Free Explore module allows students to inspect anatomical models without following a fixed lesson order. This module is designed for independent exploration and review. Students can focus on the anatomical structures they want to examine and observe them from different angles.

The module supports direct interaction with selected 3D structures. When students point to or interact with anatomical parts, the system can show the name or related information of the selected structure. This helps students connect anatomical terminology with the visual model.

Free Explore complements the AI Guided Learning module by giving students more control over the learning process. After receiving guided explanations, students can use this module to revisit the models and strengthen their spatial understanding.

6.4.3. AI Chat Module

The AI Chat module allows students to ask anatomy-related questions inside the application. It was implemented as a support module for situations where students need additional explanation beyond the prepared lesson content.

Students can submit questions through text or voice. The generated answer is shown on the interface, and it can also be read aloud depending on the selected voice settings. This makes the module useful in VR, because students can continue looking at the anatomical model while listening to the explanation.

The module is intended to support learning, not to replace teachers or official course materials. Its role is to provide quick clarification, help students review topics, and make the learning experience more interactive.

6.4.4. Quiz Module

The Quiz module was implemented to help students evaluate their understanding after the learning and exploration stages. The questions are related to the selected anatomy topics included in the current project scope, especially the musculoskeletal and circulatory system contents.

After completing a quiz, students can see their overall performance and review how many mistakes they made in each anatomical topic. The result screen also allows students to understand their topic-based performance through scores or percentage-based feedback. This helps students identify which subjects they understood well and which subjects require further review.

This feature is important for the educational purpose of the application because it connects assessment with learning awareness. Instead of only showing a final score, the system gives students a clearer view of their weak areas.

As a future improvement, the quiz results can be directly connected to the AI Guided

Learning module. If a student's success percentage in a topic is below a defined threshold, a review button can be displayed next to that topic. By selecting this button, the student can be redirected to the related AI Guided Learning content and repeat the topic immediately. This would create a stronger connection between assessment and personalized review.

6.5. Experimental Design

The project includes an experimental design to examine whether the interaction style of the avatar affects the learning experience in a VR anatomy environment. Since the avatar guides students and presents explanations, its communication style may influence student engagement, comfort, motivation, and perceived learning support.

Two experimental conditions were designed for this purpose. Both conditions use the same VR Anatomy application, modules, anatomical content, and core system functions. The difference between them is limited to the level of personalization, the language style, and the avatar presentation. This design makes it possible to focus on the effect of interaction style without changing the main learning content.

At the current stage of the project, only the first experimental condition has been completed with high school students. The second condition is planned as a follow-up study. Therefore, this report presents the completed first condition and the planned second condition, but it does not claim a completed comparison between them.

6.5.1. Experiment 1: Neutral Interaction Condition

Experiment 1 was implemented as the neutral version of the learning experience. In this condition, the system does not collect or use the student's name. The avatar presents the content with formal instructional language and has a formal visual style.

This condition represents the baseline version of the application. It was used to observe how students interact with the core VR Anatomy system without name-based personalization or a friendlier communication style.

Experiment 1 has been completed with high school students. The feedback and observations collected from this condition are used as the preliminary evaluation of the current project stage.

6.5.2. Experiment 2: Personalized Interaction Condition

Experiment 2 is planned as the personalized version of the learning experience. In this condition, the system will collect the student's name and use it during the interaction. The avatar will use friendlier and more conversational language, and its visual style will be more casual compared to Experiment 1.

The purpose of this condition is to examine whether personalization and a warmer communication style can make the learning experience feel more engaging and supportive. Since the content and modules remain the same, the comparison will focus on the effect of interaction style.

Experiment 2 has not been conducted yet. After it is completed, its results can be compared with Experiment 1 in terms of engagement, usability, comfort, motivation, and perceived learning

support.

6.7.Evaluation

The evaluation stage of the project focused on observing how high school students interacted with the VR Anatomy application and whether the system supported their understanding of selected anatomy topics. Since only the first experimental condition has been completed so far, the current evaluation is considered a preliminary classroom evaluation based on Experiment 1.

Experiment 1 was conducted with the neutral interaction condition. In this version, the system did not collect student names, used formal instructional language, and presented a formal avatar. The purpose of this evaluation was to examine the usability of the application, the clarity of the learning flow, and the students' initial learning performance after using the VR-based anatomy environment.

The evaluation included both quantitative and observational components. Quantitatively, students completed a short pre-test before using the application and a short post-test after the VR experience. Observationally, students' interaction with the VR environment, their ability to follow the learning flow, their use of the modules, and their general reactions to the application were considered.

The findings from this stage are used to understand the current strengths and limitations of the implemented system and to guide future improvements. Since Experiment 2 has not been conducted yet, this evaluation does not present a comparison between the neutral and personalized interaction conditions.

6.6.1.High School Testing Process

The first evaluation was carried out with 22 high school students, who represent the target user group of the project. Since VR Anatomy was designed to support introductory anatomy and physiology education, testing the system with high school students was important for understanding its suitability for the intended learning context.

The testing process followed three main steps. First, students completed a five-question pre- test related to the selected anatomy topics. Second, they used the VR Anatomy application under the neutral interaction condition of Experiment 1. During this stage, students experienced the implemented modules, including AI Guided Learning, free exploration, AI Chat, and quiz-based assessment. Third, students completed a five-question post-test at a similar difficulty level.

Before using the application, students were introduced to the VR environment and the basic interaction controls. This step was necessary because some students had limited or no prior experience with VR devices. During the test, students' interactions were observed to identify whether the interface, navigation, model interaction, and learning flow were understandable for the target user group.

The pre-test and post-test design allowed the project team to collect an initial quantitative measure of learning performance. At the same time, observation during the session helped

identify usability-related issues such as the need for guidance, comfort in VR, visibility of models, and clarity of interface elements.

6.6.2.Preliminary Results from Experiment 1

The recorded pre-test and post-test scores indicate a positive change in student performance after using the VR Anatomy application. The average score increased from 2.14 out of 5 in the pre- test to 2.91 out of 5 in the post-test. This corresponds to an average improvement of 0.77 points. In percentage terms, the average score increased from approximately 42.8% to 58.2%, showing an improvement of about 15.4 percentage points.

Individual score changes also showed a generally positive trend. Most students either improved their scores or maintained the same score after using the application. Specifically, 13 students improved their scores, 6 students received the same score, and 3 students received lower scores in the post-test. In addition, the highest score increased from 4 in the pre-test to 5 in the post- test.

These results suggest that the VR Anatomy application may have supported students' short- term understanding of the selected anatomy topics. The increase in the average score indicates that students were able to answer more questions correctly after interacting with the VR learning environment. However, these results should be interpreted as preliminary findings rather than final evidence of learning effectiveness.

Observation during the session also supported the quantitative results. Students were generally interested in the VR-based learning experience, and the use of 3D anatomical models helped make the content more concrete. The guided learning flow, visual highlighting, and avatar- supported explanations helped students focus on the relevant anatomical structures.

6.6.3.Limitations of the Current Evaluation

Although the preliminary results from Experiment 1 are positive, the current evaluation has several limitations. The most important limitation is that only Experiment 1 has been completed. Since Experiment 2 has not been conducted yet, the project cannot currently provide a direct comparison between the neutral interaction condition and the personalized interaction condition.

Another limitation is the size and scope of the evaluation. The test was conducted with 22 students, which provides useful early feedback but is not sufficient for a broad statistical generalization. A larger participant group would be needed to make stronger claims about the effectiveness of the system.

The post-test was applied shortly after students used the application. Therefore, the results mainly reflect short-term understanding. Long-term retention was not measured in the current evaluation. Future studies should include delayed post-tests to examine whether students remember the content after a longer period of time.

Technical factors may also have affected the evaluation. Since the application includes VR interaction, AI-supported responses, speech processing, and backend communication, the user experience can be influenced by hardware performance, internet connection quality, and backend

response time.

For these reasons, the current evaluation should be considered a preliminary classroom evaluation. It provides encouraging early results and useful feedback for future development, but further testing is needed with Experiment 2, larger participant groups, and more systematic data collection.

6.7. Conclusion

This project presented VR Anatomy, a virtual reality-based educational application designed to support anatomy learning through interactive 3D visualization, avatar-based guidance, artificial intelligence-supported explanations, and quiz-based evaluation. The main goal of the project was to make introductory anatomy education more visual, engaging, and student-centered by allowing learners to explore anatomical structures in an immersive environment.

Within the current scope, the application focused on selected topics from the musculoskeletal system and the circulatory system. The implemented system includes four main modules: AI Guided Learning, Free Explore, AI Chat, and Quiz. These modules support different stages of the learning process, including guided instruction, independent exploration, question-based clarification, and self-assessment.

The technical implementation combined a Unity-based VR application with a Python-based AI backend. Unity was used to create the 3D learning environment, VR interaction, user interface, avatar presentation, and module flow. The backend supported retrieval-based question answering, speech-to-text, and text-to-speech features. This architecture allowed the system to combine immersive visualization with intelligent learning support.

The first experimental condition was completed with high school students using the neutral interaction version of the application. The preliminary evaluation included pre-test and post-test measurements as well as observations during the VR experience. The results showed an improvement in average student performance after using the application, suggesting that VR Anatomy may support short-term understanding of selected anatomy topics.

However, the evaluation should be interpreted as preliminary rather than final evidence of learning effectiveness. Only Experiment 1 has been completed, while Experiment 2, which includes personalization, friendlier language, and a more casual avatar, is planned as a follow-up study. Therefore, a direct comparison between the two interaction conditions has not yet been completed.

Overall, the project demonstrates that VR and AI technologies can be combined to create a more interactive and supportive anatomy learning environment. VR Anatomy provides a functional prototype that can be expanded with broader anatomical content, stronger personalization, improved evaluation methods, and more advanced adaptive learning features in future work.

6.8.Future Work

Although VR Anatomy provides a functional prototype for VR-based anatomy learning, there are several directions for future improvement. These improvements can strengthen both the educational effectiveness and the technical quality of the system.

The first future direction is to complete Experiment 2 and compare it with Experiment 1. Experiment 2 will include name-based personalization, friendlier language, and a more casual avatar. After this condition is tested with students, the results can be compared with the neutral interaction condition in terms of engagement, usability, comfort, motivation, and perceived learning support. This comparison will help determine whether personalization improves the VR anatomy learning experience.

A second improvement is to conduct the evaluation with a larger participant group and a more detailed experimental design. The current evaluation provides preliminary results, but future studies should include more students, longer testing sessions, and more systematic data collection.

In addition, delayed post-tests can be used to measure whether students remember the learned content after a longer period of time.

The quiz module can also be improved by connecting quiz results directly to the learning modules. In the current version, students can see their topic-based performance after completing a quiz. In a future version, if a student's success percentage in a topic is below a defined threshold, a review button can appear next to that topic. By selecting this button, the student can be redirected to the related content in the AI Guided Learning module and repeat the topic immediately. This would create a stronger connection between assessment and personalized review.

Another future direction is to expand the anatomical content of the application. The current version focuses on selected topics from the musculoskeletal and circulatory systems. Future versions can include additional body systems such as the respiratory system, nervous system, digestive system, urinary system, and sensory organs. Expanding the content would make the application more complete and useful for broader anatomy education.

Technical improvements are also necessary for future versions. These include optimizing 3D models for better VR performance, reducing backend response time, improving speech-to-text and text-to-speech stability, and making the application more reliable on different devices. Avatar animation and lip-sync quality can also be improved to make the learning experience more natural.

Finally, future versions can include teacher-oriented features. For example, a teacher dashboard can be developed to show student progress, quiz results, frequently missed topics, and general usage statistics. This would allow teachers to use VR Anatomy not only as a student learning tool, but also as a classroom monitoring and support system.

7. Conclusion & Discussion

In this project, VR Anatomy was developed as a VR-based educational platform to support introductory anatomy learning. The main goal was to make anatomical structures easier to understand by presenting them in an immersive and interactive 3D environment. Instead of relying only on textbooks and two-dimensional visuals, students can examine selected anatomical systems, interact with models, receive explanations, and evaluate their knowledge through quizzes.

The final system includes four main modules: AI Guided Learning, Free Explore, AI Chat, and Quiz. These modules support different parts of the learning process, including guided instruction, independent exploration, question answering, and assessment. The use of avatar-based guidance and voice interaction also makes the experience more natural and suitable for a VR environment.

The project combines a Unity-based VR application with a Python-based AI backend. The backend uses Retrieval-Augmented Generation to provide anatomy-related answers based on prepared educational content. Speech-to-text and text-to-speech features allow students to communicate with the AI assistant more easily while using VR.

Overall, VR Anatomy demonstrates that VR and AI can be combined to create a more visual, interactive, and supportive anatomy learning environment.